

From Represented PGL_2 Orbit Shells to Rank-One Orientation

Reciprocity

Quadratic Residual Towers, Haar–Frobenius Comparison, and the Unramified Absolute- A_1 Theorem

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Abstract

Let

$$U_{\mathbf{Z}} = \operatorname{Spec} \mathbf{Z}[1/2, a, a^{-1}, (1-a)^{-1}], \quad \mathcal{E}_{\mathbf{Z}} = \mathcal{O}_{U_{\mathbf{Z}}}[z]/(z^2 - a).$$

For a field k with $2, \ell \in k^\times$, write $U_k = U_{\mathbf{Z}} \otimes_{\mathbf{Z}} k$ and $\mathcal{E}_k = \mathcal{E}_{\mathbf{Z}} \otimes k$. For every $r \geq 1$ we study the base change to U_k of the smooth affine truncated-unit quotient

$$\mathcal{Q}_r = \operatorname{Res}_{\mathcal{E}_k[\epsilon]/(\epsilon^r)/U_k} \mathbb{G}_m / \operatorname{Res}_{\mathcal{O}_{U_k}[\epsilon]/(\epsilon^r)/U_k} \mathbb{G}_m.$$

The first layer is the norm-one torus $\mathbb{T}$, and every subsequent reduction is a torsor under the trace-zero vector line. In a fixed enhanced ℓ -adic six-functor formalism, its compact cochains form the Frobenius orientation bimodule

$$R\pi_{\mathbb{T},!} \mathbf{Q}_\ell \simeq R\pi_{\mathbb{T},*} \mathbf{Q}_\ell \otimes \det(X_*(\mathbb{T}) \otimes \mathbf{Q}_\ell)[-1].$$

Using the fixed lift $\tilde{\gamma}_m = 1 + \varpi^m \iota(z) \in GL_2(\mathcal{O}_{U_k}[[\varpi]])$ and its class $\gamma_m \in PGL_2$, we prove that $\mathcal{Q}_r$ represents the compact positive-loop orbit sheaf of the quadratic order lattice $\mathcal{A}_r = \mathcal{O}_{U_k}[[\varpi]] + \varpi^r \mathcal{E}_k[[\varpi]]$ for $0 \leq r \leq m$. This is an isomorphism of fppf sheaves over the entire quadratic parameter space, not merely an equality of finite-field point counts. At field-valued points the quadratic-order classification gives the full support decomposition of the spherical orbital integral. The orbit stabilizers, quotient Haar masses, and Frobenius traces agree term by term.

The split and unramified-elliptic fibers are thereby unified by the Kummer local system $\mathcal{K} \simeq \det(X_*(\mathbb{T}) \otimes \mathbf{Q}_\ell)$. After Weyl normalization their generating class satisfies

$$W_{PGL_2}(z) = \frac{1 - \kappa_\ell Qz}{(1-z)(1-Qz)}, \quad W_{PGL_2}(z^{-1}) = -\kappa_\ell z d_{U_k}(W_{PGL_2}(z)).$$

For every finite depth cutoff we also prove the object-level Verdier partner identity on the discrete labelled-shell target. The separate height-cone statement is made only for the cohomological associated graded and after passing to a rational K_0 character.

We then pass from the represented PGL_2 family to a second, explicitly local level. For central-neutral standard spherical families in unramified absolute type A_1 , a classification reduction and an orbit-summed measure lemma exhaust the possible reduced-building shell lattices. The SL_2 elliptic phases retain only even or odd spheres, whereas adjoining the connected center in U_2 forces deletion of the central cocharacter direction. Nevertheless every case satisfies

$$W_G(z^{-1}) = -z(\mathfrak{o}_{G,T} \otimes d_S)W_G(z), \quad \mathfrak{o}_{G,T} = \det(X_*(T/Z(G)^\circ) \otimes \mathbf{Q}_\ell).$$

Outside the saturated PGL_2 family this is a local orbit-summed theorem, not a relative fppf representability statement. No whole affine-Springer quotient, cross-shell incidence, convolution, Rees completion, or general logarithmic skeletonization theorem is claimed.

1 Introduction

The numerical split and unramified-elliptic rank-one orbital integrals are classical. Likewise, the motivic nature of broad classes of orbital integrals was established by Hales, and transfer principles place families of fundamental-lemma integrals in motivic integration [11, 5]. Affine-Springer fundamental domains and their relation to ordinary or weighted orbital integrals have been developed systematically by Chen [2, 3, 4]. Accordingly, neither the existence of a motivic interpolation nor the rational function by itself is presented here as new. Kummer local systems have also appeared previously in orbital-integral geometry, notably in the Moy–Prasad quotient analysis of Cunningham–Hales [7].

The level isolated here is the representing-object level. The numerical benchmarks are Kottwitz’s split orbital-integral formula q^m in [12, §5.8, (5.8.4)] and his unramified elliptic fixed-vertex count $1 + (q + 1)(1 + \cdots + q^{m-1})$ in [12, §5.9, (5.9.7)]. Our contribution lies in the relative orbit construction and its descent: the stabilizer computation over arbitrary base rings in Lemma 5.4, the represented comparison in Theorem 5.5, and the quadratic orientation descent. The finite discrete target’s Verdier partner is then obtained by standard six-functor duality applied to the constructed structure morphism, as in Theorem 7.1.

The first contribution is more specific and more geometric. For the universal quadratic PGL_2 family we represent each compact positive-loop orbit shell by one smooth finite-type U -scheme $\mathcal{Q}_r$, prove the identification as an fppf orbit sheaf, and track the split/elliptic variation through the single determinant local system

$$\mathcal{K} \simeq \det(X_*(\mathbb{T}) \otimes \mathbb{Q}_\ell).$$

The vector-torsor residual tower propagates this orientation coefficient through every depth. Thus the familiar specializations $q - 1$ and $q + 1$ are fibers of one descended cohomological object rather than two separately computed formulas. The scope of the result is exactly this componentwise represented-family theorem and its orientation-compatible assembly.

The second contribution is local and structural. We classify the reduced-building data of central-neutral standard spherical families for all unramified connected reductive groups whose absolute adjoint root system is A_1 . The classification has only a saturated shell lattice or a type-preserving lattice with two elliptic embedding phases. An orbit-summed measure lemma handles the fact that one PGL_2 torus orbit may split into several SL_2 orbits. The SL_2 parity calculation changes the generating series without changing its inversion line, while the U_2 center test shows that the determinant must be taken after quotienting by $Z(G)^\circ$. This proves the unramified absolute- A_1 orientation theorem at the level of local orbital masses and their displayed K_0 interpolation.

The logical spine of the paper is therefore

quadratic residual tower $\implies$ represented PGL_2 orbit shells $\Downarrow$ Haar/Frobenius reciprocity $\implies$ finite shell skeletonization
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The first implication is geometric, the second compares Haar mass with a derived Frobenius supertrace, and the last is the fixed-map Verdier partner for the finite labelled shell map. Keeping these arrows distinct prevents the rational height-cone shadow from being mistaken for an object-level skeletonization of the whole affine-Springer fiber.

The structural extension is a second chain,

$$\begin{array}{c}
PGL_2 \text{ orientation} \implies \text{absolute-}A_1 \text{ classification} \\
\Downarrow \\
SL_2 \text{ parity and } U_2 \text{ center test} \implies \text{local rank-one orientation reciprocity}
\end{array}$$

Only the first chain contains a relative represented orbit family.

The paper is organized in the order needed to verify the two theorem layers. Section 2 fixes notation and states Theorems A and B with their different logical status. The enhanced torus calculation and residual tower come next, followed by the PGL_2 orbit-shell comparison, Haar support decomposition, and Frobenius trace. The generating function, finite labelled-shell Verdier partner, and cellular K_0 shadow are then deduced. Section B proves the classification reduction, the SL_2 parity and U_2 center tests, and the local absolute- A_1 theorem. The final section compares the result with the literature, states the limitations, and isolates the remaining categorical/logarithmic bridge.

1.1 Coefficient convention

We fix, once and for all, the constructible $\mathbf{Q}_\ell$ -linear enhanced six-functor formalism on the noetherian finite-dimensional schemes occurring below, in the sense of the enhanced operations of Laszlo–Olsson and Liu–Zheng [13, 14]. We use its symmetric monoidal stable categories $\mathcal{D}_c(S)$, internal Hom, module categories for algebra objects, relative tensor products computed by the bar construction, smooth purity, and effective finite étale descent. Finite étale pullback is conservative; descent applies to E_∞ -algebras, their module categories, and morphisms between dualizable modules. These are standing properties of the fixed formalism, not conclusions deduced from the triangulated homotopy category. We do not quantify over arbitrary enhancements.

There are two Grothendieck levels, and we keep their notation separate. Let

$$\widetilde{U}_{\mathbf{Z}} = \mathrm{Spec} \mathbf{Z}[1/2, b, b^{-1}, (1 - b^2)^{-1}] \longrightarrow U_{\mathbf{Z}}, \quad b^2 = a,$$

and set

$$\mathbb{L}_{\mathrm{mot}} = [\mathbb{A}_{U_{\mathbf{Z}}}^1], \quad \kappa_{\mathrm{mot}} = [\widetilde{U}_{\mathbf{Z}}] - [U_{\mathbf{Z}}] \quad \text{in } K_0(\mathrm{Var}_{U_{\mathbf{Z}}}).$$

After base change to U_k , compactly supported ℓ -adic realization sends these classes to

$$\mathbb{L}_\ell = [\mathbf{Q}_\ell(-1)[-2]], \quad \kappa_\ell = [\mathcal{K}] \quad \text{in } K_0(D_c^b(U_k, \mathbf{Q}_\ell)).$$

Here $\mathcal{K}$ is the anti-invariant summand defined by $R\tilde{p}_*\mathbf{Q}_\ell \simeq \mathbf{Q}_\ell \oplus \mathcal{K}$ for the base-changed quadratic cover. The torsor identity gives $\kappa_{\mathrm{mot}}^2 = 1$, hence $\kappa_\ell^2 = 1$ after realization. We put $Q = \mathbb{L}_\ell^{-1}$. Thus a geometric cover class, its Kummer-sheaf realization, and a Tate class never share the same symbol below.

1.2 Universal notation

Fix k as above. From this point onward, until an integral model is explicitly displayed, we abbreviate $U = U_k$ and $\mathcal{E} = \mathcal{E}_k$. Let $\iota : \mathcal{E} \hookrightarrow M_2(\mathcal{O}_U)$ be the regular representation in the basis $(1, z)$, so

$$\iota(z) = \begin{pmatrix} 0 & a \\ 1 & 0 \end{pmatrix}.$$

Put $G = PGL_{2,U}$ and

$$\mathbb{T} = \mathrm{Res}_{\mathcal{E}/U} \mathbb{G}_m / \mathbb{G}_m \simeq \ker(N_{\mathcal{E}/\mathcal{O}_U}), \quad \tilde{\gamma}_m = 1 + \varpi^m \iota(z), \quad \gamma_m = [\tilde{\gamma}_m].$$

Let $X_m \subset \mathrm{Gr}_G$ be the intrinsic torsor-valued affine-Springer subfunctor defined in (21); on a section represented by a loop g , its condition is $g^{-1}\gamma_m g \in L^+G$. Also put $\mathcal{A}_r = \mathcal{O}_U[[\varpi]] + \varpi^r \mathcal{E}[[\varpi]]$. All appearances of an order generated by γ_m below mean the order generated by the fixed GL_2 -lift $\tilde{\gamma}_m$.

The companion representative used in earlier calculations is

$$\gamma_{s,a}^{\text{comp}} = \begin{bmatrix} 1 & as^2 \\ 1 & 1 \end{bmatrix}.$$

For $s = \varpi^m$ it is conjugate to the integral quadratic-basis model:

$$\begin{pmatrix} 1 & 0 \\ 0 & s \end{pmatrix} \begin{pmatrix} 1 & as^2 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & s^{-1} \end{pmatrix} = 1 + s\iota(z). \quad (1)$$

The paper uses only $\gamma_m = [1 + \varpi^m \iota(z)]$; the companion form is recorded solely to identify the two presentations of the same regular class.

We reserve separate notation for the four orbital levels. At a finite-field point x , $\text{OI}_m(x)$ denotes the numerical Haar integral. The represented shell union will define the coefficient object $\mathcal{I}_m^{\text{orb}} = Rq_{m,!}^{\text{orb}} \mathbf{Q}_\ell$, its raw class is $I_m^{\text{orb}} = [\mathcal{I}_m^{\text{orb}}]$, and its Weyl-normalized class is

$$J_m^{\text{orb}} = Q^m I_m^{\text{orb}}.$$

Only the normalized coefficients enter $W_{PGL_2}(z) = \sum_{m \geq 0} J_m^{\text{orb}} z^m$. Thus a numerical orbital integral, a coefficient object, its Grothendieck class, and its normalization are never denoted by the same symbol.

Proposition 1.1 (The Kummer coefficient is strictly non-Tate). *The local system $\mathcal{K}$ on U_k does not belong to the thick tensor subcategory generated by constant Tate sheaves. Consequently every positive residual layer contains a genuinely non-Tate cohomology sheaf; the correction κ_ℓ in the normalized orbital coefficients cannot be replaced by a Tate power.*

Proof. After base change to an algebraic closure, the cover $b^2 = a$ of $\mathbb{G}_m \setminus \{1\}$ is connected and has quadratic monodromy around $a = 0$. Hence its anti-invariant summand $\mathcal{K}$ has monodromy -1 , whereas constant Tate sheaves have trivial geometric monodromy. The compact cohomology calculation in Lemma 3.3, followed by the vector-torsor Thom isomorphism (14), places a Tate twist of $\mathcal{K}$ in every $Rq_{r,!} \mathbf{Q}_\ell$ with $r \geq 1$. Weight and monodromy separate it from the Tate summand, so it cannot disappear through a Tate re-expression. $\square$

Remark 1.2. The non-Tate property of a nontrivial quadratic Kummer local system is standard. Proposition 1.1 is included only to certify that the coefficient occurring in the represented shells cannot be absorbed into a Tate normalization.

2 Main results and status

There are two theorem layers. Theorem A is relative and object-level for the universal quadratic PGL_2 family. Theorem B is a local orbit-summed theorem for the explicitly defined unramified absolute- A_1 class. We state both before introducing the formal lemmas used later and keep their logical status separate throughout.

Theorem 2.1 (Theorem A: the universal quadratic PGL_2 package). *Let k be a field with $2 \in k^\times$, let ℓ be invertible in k , and put*

$$U_k = U_{\mathbf{Z}} \otimes_{\mathbf{Z}} k = \text{Spec } k[a, a^{-1}, (1-a)^{-1}], \quad \mathcal{E}_k = \mathcal{O}_{U_k}[z]/(z^2 - a).$$

For the universal family $\gamma_m = [1 + \varpi^m \iota(z)] \in PGL_2(\mathcal{O}_{U_k}((\varpi)))$ the following statements hold.

- (i) *The truncated-unit fppf quotients $\mathcal{Q}_r$ are represented by smooth affine U -group schemes. One has $\mathcal{Q}_1 = \mathbb{T}$, the norm-one torus, and $\mathcal{Q}_r \rightarrow \mathcal{Q}_{r-1}$ is a torsor under the trace-zero vector line. In the enhanced bimodule category,*

$$R\pi_{\mathbb{T},!} \mathbf{Q}_\ell \simeq R\pi_{\mathbb{T},*} \mathbf{Q}_\ell \otimes \det(X_*(\mathbb{T}) \otimes \mathbf{Q}_\ell)[-1].$$

- (ii) For every $0 \leq r \leq m$, the positive-loop orbit sheaf of the order lattice $\mathcal{A}_r = \mathcal{O}_{U_k}[[\varpi]] + \varpi^r \mathcal{E}_k[[\varpi]]$ is represented, and its orbit morphism is a monomorphism of fppf sheaves

$$\mathcal{Q}_r \xrightarrow{\sim} L^+ \mathbb{T} \cdot [\mathcal{A}_r] \longrightarrow X_m.$$

At every field-valued point, the classes $[\mathcal{A}_r]$, $0 \leq r \leq m$, are exactly the representatives of $T(F) \backslash X_m$.

- (iii) With Haar measure normalized by $\text{vol}(PGL_2(\mathcal{O}_F)) = \text{vol}(T(\mathcal{O}_F)) = 1$ and quotient measure defined by Weil integration, the mass of the r th double coset is

$$[T(\mathcal{O}_F) : A_r^\times / \mathcal{O}_F^\times].$$

Over a finite residue field this is both $\# \mathcal{Q}_r(k)$ and the geometric-Frobenius derived supertrace of its compact coefficient.

- (iv) After Weyl-discriminant normalization, the compact-orbital generating class is

$$\begin{aligned} J_m^{\text{orb}} &= Q^m I_m^{\text{orb}}, \\ W_{PGL_2}(z) &= \sum_{m \geq 0} J_m^{\text{orb}} z^m = \frac{1 - \kappa_\ell Qz}{(1 - z)(1 - Qz)}, \\ W_{PGL_2}(z^{-1}) &= -\kappa_\ell z d_{U_k}(W_{PGL_2}(z)). \end{aligned}$$

where $I_m^{\text{orb}} = [Rq_{m,!}^{\text{orb}} \mathbf{Q}_\ell]$ is the raw compact-orbital coefficient, $\kappa_\ell = [\det(X_*(\mathbb{T}) \otimes \mathbf{Q}_\ell)] = [\mathcal{K}]$, and d_{U_k} is the dimension-normalized coefficient duality of (38).

- (v) For every finite cutoff N , the actual labelled-shell map $\rho_N : \mathcal{Y}_N^{\text{orb}} \rightarrow U_k \times \Lambda_N$ defines $\text{Sk}_{N,!} = R\rho_{N,!}$ and $\text{Sk}_{N,*} = R\rho_{N,*}$ independently, and

$$\mathbb{D}_{\text{cell},N} \text{Sk}_{N,!} \simeq \text{Sk}_{N,*} \mathbb{D}_{\mathcal{Y}_N^{\text{orb}}}$$

naturally on the full bounded constructible category.

- (vi) The associated-graded shell coefficients have canonical height $s = m - r + 1$. Their de-Tated height character is represented by the auxiliary cellular envelope $(1 - \kappa_\ell) \mathbf{1}_C + \kappa_\ell \mathbf{1}_F$; the variable u^s records the twist (s), and the specialization $u \mapsto Q$ recovers the actual ordinary K_0 class. The resulting cellular reciprocity is a rational K_0 statement, not an object-level sheaf on the incidence-bearing cone.

Proof. Part (i) is Theorems 3.2 and 4.1, with the enhanced algebra model supplied by Lemma 3.1. Part (ii) is Lemma 5.1, Lemma 5.4, and Theorem 5.5. Part (iii) is Lemma 5.6 and Corollary 5.8, including (31) and (33). Part (iv) is Theorem 5.10. Parts (v) and (vi) are Theorems 7.1 and 7.3, respectively. $\square$

Definition 2.2 (Unramified absolute- A_1 pair). Let F be a non-Archimedean local field with finite residue field of odd characteristic. An *unramified absolute- A_1 pair* (G, K) consists of a connected reductive F -group G which is quasi-split over F , split over an unramified extension, and has absolutely simple adjoint group of type A_1 , together with a hyperspecial subgroup $K = \mathcal{G}(\mathcal{O}_F)$ for a connected reductive $\mathcal{O}_F$ -model $\mathcal{G}$. The adjective A_1 refers to the absolute adjoint root system.

Definition 2.3 (Central-neutral standard spherical family). Let $p : G \rightarrow G_{\text{ad}}$, let $\nu : G \rightarrow G_{\text{ab}} = G/G_{\text{der}}$, and let $K_{\text{ad}} = \mathcal{G}_{\text{ad}}(\mathcal{O}_F)$, and let $T = G_{\gamma_m}$ with root image $T_{\text{rt}} \subset G_{\text{ad}}$. A family γ_m , $m \geq 0$, is *central-neutral standard spherical* if:

- (a) T_{rt} is split or unramified elliptic and its adjoint quadratic order is $A_m = \mathcal{O}_F + \varpi^m \mathcal{O}_E$ for the corresponding split or unramified quadratic algebra E/F ;
- (b) its fixed set in the reduced building is the radius- m tube about the split apartment or the radius- m ball about the elliptic fixed vertex;
- (c) $\nu(\gamma_m)$ is independent of m , belongs to $\nu(K)$, and

$$\mathbf{1}_K(g^{-1}\gamma_m g) = \mathbf{1}_{K_{\text{ad}}}(p(g)^{-1}p(\gamma_m)p(g)) \quad (g \in G(F)); \quad (2)$$

- (d) the Haar measures satisfy $\text{vol}_G(K) = 1$ and $\text{vol}_T(T(F)_{\text{max,cpt}}) = 1$, and the Weyl discriminant has valuation $2m$.

These conditions are hypotheses, not automatic properties of an arbitrary unramified regular family.

For such a centralizer put

$$\mathfrak{o}_{G,T} = \det(X_*(T/Z(G)^\circ) \otimes \mathbf{Q}_\ell). \quad (3)$$

Theorem 2.4 (Theorem B: unramified absolute- A_1 orientation reciprocity). *Let (G, K) be an unramified absolute- A_1 pair and let γ_m be a central-neutral standard spherical family. Let S/k be a smooth connected parameter base over a finite field k , with $\ell \neq \text{char } k$, carrying the quadratic cameral cover of T_{rt} , and let d_S be dimension-normalized coefficient duality. The classification in Section B is exhaustive: the shell lattice is either saturated or type-preserving, and in the type-preserving elliptic case it has exactly two embedding phases. The local Haar coefficients are given by the corresponding orbit-summed shell interpolation $I_{G,m}$.*

With $J_{G,m} = Q^m I_{G,m}$ and $W_G(z) = \sum_{m \geq 0} J_{G,m} z^m$, one has

$$\boxed{W_G(z^{-1}) = -z(\mathfrak{o}_{G,T} \otimes d_S)W_G(z)}. \quad (4)$$

The multiplier is independent of the central isogeny, vertex-color lattice, and elliptic embedding phase. Outside the saturated PGL_2 case the interpolation is a rational K_0 class whose Frobenius traces are the actual local orbital coefficients; no relative orbital R_1 -push-forward or fppf orbit-family theorem is asserted.

Proof. This is Theorems B.8 and B.9. The classification reduction is Lemma B.3, the measure comparison is Lemma B.4, the type-preserving calculation is Proposition B.6, and the quotient by $Z(G)^\circ$ is forced by Proposition B.7. $\square$

For later reference, the logical status of the two theorem packages is as follows.

Residual tower and torus bimodule. Proved at scheme level and in the selected enhanced coefficient category; no global orbital skeleton is inferred from it.

Positive-loop shell comparison. Proved for represented fppf orbit sheaves; this is not promoted to a stratification of X_m or $[T \setminus X_m]$.

Orbital mass and reciprocity. Proved as a quotient-measure identity, a Frobenius-trace formula, and a rational K_0 identity; no ungraded cone sheaf or convolution is asserted.

Finite labelled-shell partner. Proved at object level on a zero-dimensional labelled skeleton; cross-shell incidence and a Rees action remain separate problems.

Height-cone shadow. Proved after applying the de-Tated graded character and passing to K_0 ; this does not supply an incidence-bearing object on the height cone.

Absolute- A_1 extension. Proved for Definitions 2.2 and 2.3 as a classification and equality of local orbit-summed Haar masses, assembled into an explicit rational K_0 interpolation. Only its saturated PGL_2 member has the relative represented-shell geometry of Theorem A.

2.1 Proof routes

The arithmetic core may be read independently of most of the formal framework. The shortest proof route is:

1. the enhanced torus model and residual tower, Theorem 3.2 and Section 4;
2. quadratic orders and represented orbit sheaves, Section 5;
3. quotient measure, trace, and orbital reciprocity, Corollary 5.8 and Theorem 5.10;
4. the finite shell Verdier partner and the separate height-character shadow, Section 7.

The short cellular section at the end is used only for the rational K_0 shadow; it is not a prerequisite for the orbit-shell or Haar statements. Theorem B has a separate route: finite central lifting, classification of the type image in the reduced building, orbit-summed measure transfer, the two SL_2 elliptic phases, and the U_2 connected-center test. None of those steps is used to strengthen the relative representability assertion of Theorem A.

3 The enhanced torus orientation bimodule

Lemma 3.1 (Enhanced Kummer–Koszul algebra and its bimodule dual). *Let $p : S' \rightarrow S$ be a finite étale cover splitting the d -dimensional torus T , let $\pi' : T_{S'} \rightarrow S'$, and put*

$$V_T = X^*(T_{S'}) \otimes_{\mathbf{Z}} \mathbf{Q}_\ell(-1).$$

In the fixed enhanced six-functor formalism of the coefficient convention there is a functorial equivalence of augmented E_∞ -algebras

$$K(V_T) := \mathrm{Sym}(V_T[-1]) \xrightarrow{\sim} R\pi'_* \mathbf{Q}_\ell. \quad (5)$$

Because the coefficient category is $\mathbf{Q}_\ell$ -linear, the underlying object is

$$K(V_T) \simeq \bigoplus_{i=0}^d \bigwedge^i V_T[-i], \quad (6)$$

with exterior multiplication. Projection to the top exterior power defines a perfect Frobenius pairing and a functorial equivalence of $K(V_T)$ -bimodules

$$K(V_T)^\vee \xrightarrow{\sim} K(V_T) \otimes (\det V_T)^\vee[d]. \quad (7)$$

The left and right actions in this equivalence are the two actions induced by the E_∞ multiplication, and evaluation is multiplication followed by projection to $\det V_T[-d]$. These maps, including evaluation and coevaluation, are functorial for automorphisms of the character lattice. Consequently their pullbacks to every level of the Čech nerve of S'/S satisfy the descent cocycle.

Proof. For a character $\chi : T_{S'} \rightarrow \mathbb{G}_m$, pullback of the universal Kummer class gives a degree-one map $X^*(T_{S'}) \otimes \mathbf{Q}_\ell(-1)[-1] \rightarrow R\pi'_* \mathbf{Q}_\ell$. Additivity in χ and the universal property of the free commutative algebra extend it to the augmented E_∞ -algebra map in (5). For a split rank-one torus, the Kummer pro-étale tower computes this map by the two-term continuous Koszul resolution, compatibly with cup products and the augmentation. It is therefore a quasi-isomorphism of E_∞ -algebras in rank one. The symmetric-monoidal Künneth equivalence identifies the rank- d map with the tensor product of these rank-one maps. The forgetful functor from E_∞ -algebras to the underlying stable category is conservative, and the bounded constructible t -structure detects equivalences. Thus this is an object-level E_∞ -equivalence, not an inference made only after passing to cohomology or K_0 , and its underlying object is (6).

Exterior multiplication followed by the top-wedge projection pairs $\bigwedge^i V_T$ perfectly with $\bigwedge^{d-i} V_T$. The resulting adjoint map is (7); associativity and graded commutativity give the Frobenius balance identity

$$\beta(xy, z) = \beta(x, yz), \quad \beta : K(V_T) \otimes K(V_T) \longrightarrow \det V_T[-d],$$

including the Koszul signs. Hence the adjoint is simultaneously left- and right-linear. The standard perfect-pairing evaluation-coevaluation composites give the triangle identities. Every construction used only characters, Kummer pullback, multiplication, and the top exterior projection. It is therefore functorial for lattice isomorphisms. On the Čech nerve the two pullbacks are related by the descent automorphism of $X^*(T)$, and functoriality makes the triple-overlap comparison the corresponding product. This is exactly the descent cocycle. $\square$

Theorem 3.2 (Torus Frobenius bimodule). *Let S be a noetherian finite-dimensional scheme on which ℓ is invertible. Work in the fixed enhanced ℓ -adic formalism declared in the coefficient convention; write $\mathcal{D}_c(S)$ for its stable category and $D_c^b(S, \mathbf{Q}_\ell)$ for the bounded constructible homotopy category. Let $\pi : T \rightarrow S$ be a smooth separated torus of constant relative dimension d , split by a finite étale cover. Let*

$$\Lambda_T := X_*(T) \otimes_{\mathbf{Z}} \mathbf{Q}_\ell$$

be the lisse $\mathbf{Q}_\ell$ -sheaf obtained from the étale sheaf of cocharacters, and write

$$A_T^* = R\pi_* \mathbf{Q}_\ell, \quad A_T^! = R\pi_! \mathbf{Q}_\ell, \quad \mathfrak{o}_T = \det(\Lambda_T).$$

Cup product is taken in the enhancement and makes A_T^ an E_∞ -algebra. Put $A_T^e = A_T^* \otimes^L (A_T^*)^{\text{op}}$, and let $(-)^{\vee} = R\mathcal{H}om_{\mathbf{Q}_\ell}(-, \mathbf{Q}_\ell)$ denote coefficient-linear duality, internal to $D_c^b(S, \mathbf{Q}_\ell)$, not Verdier duality on S . Write $D(A_T^e)$ for the homotopy category of constructible A_T^e -modules in $\mathcal{D}_c(S)$. The projection-formula cup action makes $A_T^!$ an invertible bimodule for the relative tensor product over A_T^* , as the formulas below show. Relative tensors are computed in the ambient derived module category admitting the bar construction; the displayed bimodule, its inverse, and their relative tensor products remain constructible. This is Morita invertibility for $\otimes_{A_T^*}^L$, not perfectness as an A_T^e -module for $\otimes_{A_T^e}^L$. All tensor products in this theorem and its proof are derived, even when the superscript L is suppressed for readability. Exterior multiplication and relative Poincaré duality give a canonical equivalence there*

$$\boxed{A_T^! \simeq A_T^* \otimes \mathfrak{o}_T[-d]}. \quad (8)$$

Equivalently,

$$A_T^! \simeq (A_T^*)^{\vee}(-d)[-2d]. \quad (9)$$

The coefficient-duality evaluation is cup product followed by the relative trace

$$A_T^* \otimes A_T^! \longrightarrow \mathbf{Q}_\ell(-d)[-2d].$$

Under (8), bimodule evaluation and coevaluation are multiplication together with the ordinary evaluation and coevaluation of $\mathfrak{o}_T[-d]$. They satisfy the triangle identities and exhibit $A_T^!$ as an invertible dualizing A_T^* -bimodule, with explicit inverse

$$(A_T^!)^{-1} = A_T^* \otimes \mathfrak{o}_T^\vee[d]. \quad (10)$$

Here the Morita evaluation is the A_T^* -linear map $A_T^! \otimes_{A_T^*}^L (A_T^!)^{-1} \rightarrow A_T^*$; it is distinct from the coefficient pairing displayed above. The determinant line is therefore forced before taking K_0 .

More explicitly, if $L = A_T^* \otimes \mathfrak{o}_T[-d]$ and $L^{-1} = A_T^* \otimes \mathfrak{o}_T^\vee[d]$, the duality maps have domains and codomains

$$\begin{aligned} \mathrm{ev}_A : L \otimes_{A_T^*}^L L^{-1} &\longrightarrow A_T^*, \\ \mathrm{coev}_A : A_T^* &\longrightarrow L^{-1} \otimes_{A_T^*}^L L. \end{aligned}$$

The composites $L \rightarrow L \otimes_A L^{-1} \otimes_A L \rightarrow L$ and $L^{-1} \rightarrow L^{-1} \otimes_A L \otimes_A L^{-1} \rightarrow L^{-1}$ are the identity.

The duality $(-)^{\vee}$ in this statement is not raw Verdier duality $\mathbb{D}_S^{\mathrm{raw}} = R\mathcal{H}om_S(-, \omega_S)$. In particular, no dimension-normalized coefficient involution is being used in (8); the normalized operation d_{U_k} used in the generating series is defined separately in (38) below.

Proof. The cup action $A_T^* \otimes A_T^! \rightarrow A_T^!$ is the projection-formula form of the usual action of ordinary cochains on compactly supported cochains. After the splitting cover, put $V = X^*(T) \otimes \mathbf{Q}_\ell(-1)$. The Kummer pro-étale tower gives the Koszul model for the continuous cochains of the free rank- d Tate lattice precisely as constructed in Lemma 3.1. The Koszul resolution is self-dual, with determinant its top exterior generator; therefore it gives at chain level, as a bimodule,

$$(A_T^*)^\vee \simeq A_T^* \otimes (\det V)^\vee[d] \simeq A_T^* \otimes \mathfrak{o}_T(d)[d].$$

Here the determinant calculation, including every dual and twist, is

$$\det V = \det(X^*(T) \otimes \mathbf{Q}_\ell) \otimes \mathbf{Q}_\ell(-d) = \mathfrak{o}_T^\vee(-d), \quad (\det V)^\vee = \mathfrak{o}_T(d).$$

Its cohomology is the familiar exterior algebra $\bigwedge^\bullet V$, so the chain-level determinant is compatible with the top-wedge pairing. Let $K(V)$ denote this finite Koszul model. Contraction of exterior powers, followed by projection to the top exterior generator, gives the coefficient pairing $K(V) \otimes K(V)^\vee \rightarrow \mathbf{Q}_\ell$; exterior multiplication gives the $K(V)$ -module map realizing (8). Tensoring that map with the ordinary evaluation $\mathfrak{o}_T \otimes \mathfrak{o}_T^\vee \rightarrow \mathbf{Q}_\ell$ gives

$$(A_T^* \otimes \mathfrak{o}_T[-d]) \otimes_{A_T^*}^L (A_T^* \otimes \mathfrak{o}_T^\vee[d]) \longrightarrow A_T^*.$$

Insertion of the coevaluation of $\mathfrak{o}_T$ gives its inverse-direction coevaluation. The two composites are the identity because they are the usual evaluation–coevaluation composites of an invertible line; this checks the triangle identities in $D(A_T^e)$. Relative Poincaré duality gives (9); substitution yields (8). Its compatibility with the left and right A_T^* -actions is the projection-formula identity for cup product and the relative trace: after pullback to the splitting cover it is exactly the $K(V)$ -balanced top-wedge pairing of Lemma 3.1, and the splitting pullback is conservative. Concretely, the relative trace pairing satisfies

$$\mathrm{tr}((ab)c) = \mathrm{tr}(a(bc))$$

with the graded signs; this is the A_T^e -balance condition that makes the adjoint map a bimodule morphism. After replacing a finite splitting cover by a finite étale Galois refinement, every

Koszul map, multiplication, trace, evaluation, and coevaluation above is equivariant for the splitting group. The Čech cocycle is the lattice-functorial cocycle proved in that lemma. Effective finite étale descent for E_∞ -algebras and their module categories therefore descends the enhanced bimodule equivalence to S . Compatibility with exterior multiplication makes it a bimodule, not merely a coefficient, equivalence. $\square$

Lemma 3.3 (Compact cohomology of a torus). *Let $\pi : T \rightarrow S$ be a smooth d -dimensional torus split by a finite étale cover, and put $\Lambda = X_*(T) \otimes \mathbf{Q}_\ell$. Then*

$$R^{2d-i}\pi_!\mathbf{Q}_\ell \simeq \bigwedge^i \Lambda (i-d), \quad 0 \leq i \leq d. \quad (11)$$

In particular

$$R^d\pi_!\mathbf{Q}_\ell \simeq \det \Lambda, \quad R^{2d}\pi_!\mathbf{Q}_\ell \simeq \mathbf{Q}_\ell(-d).$$

For $d = 1$ its ℓ -adic Grothendieck class is $\mathbb{L}_\ell - [\det \Lambda]$.

Proof. After the splitting cover, $R^i\pi_*\mathbf{Q}_\ell \simeq \bigwedge^i X^*(T) \otimes \mathbf{Q}_\ell(-i)$. Smooth Poincaré duality identifies $R^{2d-i}\pi_!\mathbf{Q}_\ell$ with the dual of this local system twisted by $(-d)$. The formula follows, and it descends because all identifications are equivariant for the splitting group. $\square$

4 The universal quadratic residual filtration

This section constructs the finite residual coefficient objects. The next section identifies them with compact positive-loop orbits in an actual equal-characteristic PGL_2 affine-Springer locus. Neither section constructs a sheaf on the whole valued-field quotient, an infinite completed Rees object, or an orbital convolution kernel.

The residual tower is first defined over the arithmetic parameter space $U_{\mathbf{Z}}$. Put

$$\mathcal{E}_{\mathbf{Z}} = \mathcal{O}_{U_{\mathbf{Z}}}[z]/(z^2 - a), \quad \mathcal{E}_{\mathbf{Z}}^- = \ker(\mathrm{Tr}_{\mathcal{E}_{\mathbf{Z}}/\mathcal{O}_{U_{\mathbf{Z}}}}).$$

For $r \geq 1$ define the integral fppf quotient

$$\mathcal{Q}_{r,\mathbf{Z}} = \frac{\mathrm{Res}_{\mathcal{E}_{\mathbf{Z}}[\epsilon]/(\epsilon^r)/U_{\mathbf{Z}}} \mathbb{G}_m}{\mathrm{Res}_{\mathcal{O}_{U_{\mathbf{Z}}}[\epsilon]/(\epsilon^r)/U_{\mathbf{Z}}} \mathbb{G}_m}. \quad (12)$$

This is represented by a smooth affine $U_{\mathbf{Z}}$ -group scheme: after the finite étale cover splitting $\mathcal{E}_{\mathbf{Z}}$, it is a truncated unit group, and effective affine descent applies.

Now let k be a field with $2, \ell \in k^\times$, base change to U_k , and write $U = U_k$, $\mathcal{E} = \mathcal{E}_k$, $\mathcal{E}^- = \mathcal{E}_{\mathbf{Z}}^- \otimes k$, and $\mathcal{Q}_r = \mathcal{Q}_{r,\mathbf{Z}} \otimes k$. Write $q_r : \mathcal{Q}_r \rightarrow U$ and $H_r = Rq_{r,!}\mathbf{Q}_\ell$. The quotient in (12) means the fppf sheaf quotient; all subsequent ℓ -adic identities live over $U = U_k$.

Theorem 4.1 (Quadratic residual tower and its orientation coefficient). *Let $\mathbb{T} = \mathcal{Q}_1$, let $A = R\pi_{\mathbb{T},*}\mathbf{Q}_\ell$, and let $\omega_A = R\pi_{\mathbb{T},!}\mathbf{Q}_\ell$.*

- (i) *The fppf quotient $\mathcal{Q}_1$ is canonically the norm-one torus $\ker(N_{\mathcal{E}/\mathcal{O}_U} : \mathrm{Res}_{\mathcal{E}/U} \mathbb{G}_m \rightarrow \mathbb{G}_m)$.*
- (ii) *For $r \geq 2$, reduction modulo ϵ^{r-1} is an fppf torsor*

$$p_r : \mathcal{Q}_r \longrightarrow \mathcal{Q}_{r-1} \quad \text{under } \mathbb{V}(\mathcal{E}^-). \quad (13)$$

Consequently

$$H_r \simeq \omega_A(-(r-1))[-2(r-1)]. \quad (14)$$

- (iii) Here (m) denotes the ordinary Tate twist used for the chosen Weyl normalization; its class is Q^m in the convention of Section 6. After this twist and the even relative-dimension shift,

$$H_r(m)[2r-2] \simeq \omega_A(m-r+1) \simeq A \otimes \mathcal{K}[-1](m-r+1). \quad (15)$$

Thus the residual coefficient height is $s = m - r + 1$.

- (iv) The motivic and ℓ -adic class identities are, respectively,

$$[\mathcal{Q}_r, \mathbf{Z}] = (\mathbb{L}_{\text{mot}} - \kappa_{\text{mot}}) \mathbb{L}_{\text{mot}}^{r-1} \quad \text{in } K_0(\text{Var}_{U_{\mathbf{Z}}}), \quad (16)$$

and

$$[Rq_{r,!} \mathbf{Q}_{\ell}] = (\mathbb{L}_{\ell} - \kappa_{\ell}) \mathbb{L}_{\ell}^{r-1} \quad \text{in } K_0(D_c^b(U_k, \mathbf{Q}_{\ell})). \quad (17)$$

Proof. Because 2 is invertible, the quadratic algebra is étale and its trace-zero summand is a line bundle. The morphism $x \mapsto x/\sigma(x)$ identifies the first quotient with the norm-one torus after the splitting cover; the identification descends and proves (i). For $r \geq 2$, the kernel of reduction is

$$\frac{1 + \epsilon^{r-1} \mathcal{E}}{1 + \epsilon^{r-1} \mathcal{O}_U} \simeq \mathcal{E}/\mathcal{O}_U \simeq \mathcal{E}^-.$$

This calculation is equivariant for the quadratic descent action, so the reduction morphism is the asserted vector-line torsor, proving (13). Compact homotopy invariance gives $Rp_{r,!} \mathbf{Q}_{\ell} \simeq \mathbf{Q}_{\ell}(-1)[-2]$. Iteration proves (14).

The quadratic deck involution acts by -1 on $X_*(\mathbb{T})$, so $\det(X_*(\mathbb{T}) \otimes \mathbf{Q}_{\ell})$ is precisely the sign local system $\mathcal{K}$. Theorem 3.2 in dimension one therefore gives $\omega_A \simeq A \otimes \mathcal{K}[-1]$, proving (15). Finally the vector-torsor tower over $U_{\mathbf{Z}}$ gives (16), using $[\mathbb{T}_{\mathbf{Z}}] = \mathbb{L}_{\text{mot}} - \kappa_{\text{mot}}$; the latter follows by subtracting the quadratic boundary $\tilde{U}_{\mathbf{Z}}$ from the split conic compactification. Compactly supported realization gives (17), and Lemma 3.3 identifies the realized first layer as $\mathbb{L}_{\ell} - \kappa_{\ell}$. $\square$

Remark 4.2 (The residual filtration before orbital comparison). For a fixed m , the disjoint union

$$\mathcal{B}_m = U \amalg \coprod_{1 \leq r \leq m} \mathcal{Q}_r, \quad F_{\leq r} \mathcal{B}_m = U \amalg \coprod_{1 \leq j \leq r} \mathcal{Q}_j$$

is a finite split filtration by open-and-closed pieces, with positive coefficient layers H_r . Theorem 5.5 will identify each piece with an actual compact orbit shell. The split union itself still creates no geometric transition maps between different orbital depths. In particular the reduction morphisms $\mathcal{Q}_r \rightarrow \mathcal{Q}_{r-1}$ are not a Rees action, and no inverse system or completion in m is asserted.

Proposition 4.3 (The residual Kummer coefficient). *The actual compact coefficient complex of the first residual layer is ω_A . With $A^e = A \otimes^L A^{\text{op}}$ and $D(A^e)$ interpreted in the enhanced sense fixed in Theorem 3.2, one has*

$$\omega_A \simeq A \otimes \mathcal{K}[-1].$$

This is an object-level Frobenius-bimodule identity for the residual torus. After Theorem 5.5 it is also the vertical coefficient identity for the first actual compact orbit shell. By itself it is not a skeletonization theorem or a completed Rees self-duality.

5 From residual quotients to actual PGL_2 compact orbital shells

We now work in equal characteristic. The purpose of this section is to close one precise geometric gap: the schemes $\mathcal{Q}_r$ will be identified with positive-loop torus orbits inside an actual group affine-Springer locus. The affine Grassmannian is used in its standard fpqc-sheaf sense; its representability and basic orbit geometry are recalled in [16, 18]. We do not identify the whole quotient stack of the affine-Springer locus with a finite-type scheme.

Put $\mathcal{O}_R = R[[\varpi]]$ and $F_R = R((\varpi))$ for a U -algebra R . In the basis $(1, z)$ of $\mathcal{E}$, use the regular representation

$$\iota(z) = \begin{pmatrix} 0 & a \\ 1 & 0 \end{pmatrix}, \quad \iota(z)^2 = a. \quad (18)$$

Let $G = PGL(\mathcal{E}) \simeq PGL_{2,U}$ and let

$$\gamma_m = [1 + \varpi^m \iota(z)] = \left[\begin{pmatrix} 1 & a\varpi^m \\ \varpi^m & 1 \end{pmatrix} \right] \in G(\mathcal{O}_U((\varpi))), \quad m \geq 0. \quad (19)$$

Its determinant is $1 - a\varpi^{2m}$. This is a unit for $m > 0$, and for $m = 0$ the same statement is exactly why $(1 - a)^{-1}$ was included in U . By (1), this is the same regular semisimple conjugacy family as the companion presentation, now written in the integral quadratic basis used throughout the orbit-sheaf proof.

The centralizer of γ_m is the unramified torus

$$\mathbb{T} = (\mathrm{Res}_{\mathcal{E}/U} \mathbb{G}_m) / \mathbb{G}_m. \quad (20)$$

Write $\mathrm{Gr}_G = LG/L^+G$ as an fpqc quotient sheaf. Intrinsically, an R -point is a G -torsor $\mathcal{P}$ on $D_R = \mathrm{Spec} R[[\varpi]]$ together with a trivialization on $D_R^\times = \mathrm{Spec} R((\varpi))$. The trivialization lets γ_m act on $\mathcal{P}|_{D_R^\times}$. Define the group affine-Springer subfunctor by

$$X_m(R) = \{(\mathcal{P}, \alpha) : \alpha^{-1} \gamma_m \alpha \text{ extends to an automorphism of } \mathcal{P} \text{ on } D_R\}. \quad (21)$$

When a section is represented by $g \in LG(R)$ this is the familiar condition $g^{-1} \gamma_m g \in L^+G(R)$. The torsor formulation is required over a general base: we do not assume that a PGL_2 loop admits a GL_2 lift after a cover pulled back from $\mathrm{Spec} R$. Affine-Springer fibers and their relation to orbital integrals provide the broader context in [10].

Lemma 5.1 (Projective-lattice descent and the liftable locus). *Let R be a U -algebra.*

- (i) *The condition in (21) is invariant under isomorphism of the pair $(\mathcal{P}, \alpha)$ and is compatible with fpqc base change. Hence it defines a sub-fpqc-sheaf of Gr_G without choosing a GL_2 lift.*
- (ii) *Suppose $(\mathcal{P}, \alpha)$ admits a lift to a rank-two projective $R[[\varpi]]$ -module L with a trivialization over $R((\varpi))$. Then membership in $X_m(R)$ is equivalent to*

$$\tilde{\gamma}_m L = L,$$

and also to stability under the order

$$\mathcal{A}_{m,R} = R[[\varpi]] + \varpi^m (\mathcal{E} \otimes_{\mathcal{O}_U} R)[[\varpi]] = R[[\varpi]][\tilde{\gamma}_m].$$

Tensoring L by a line bundle, or changing a local matrix lift by a scalar, does not change either condition.

- (iii) *Every positive-loop orbit section $u[\mathcal{A}_r]$ used below has the explicit lift $u\mathcal{A}_{r,R}$. Every field-valued point of Gr_G is liftable as well. Thus the arbitrary-base orbit theorem and the field-valued quadratic-order classification both lie in the liftable locus, although no lifting assertion is made for an arbitrary R -point of the whole affine Grassmannian.*

Proof. The Beauville–Laszlo torsor description of the affine Grassmannian is fpqc functorial. Extension of a generic automorphism across D_R is invariant under descent isomorphisms, proving (i).

For (ii), work fpqc locally on D_R where the projective module is free and choose a matrix representative g . Put $h = g^{-1}\tilde{\gamma}_m g$. If the class of h is integral in PGL_2 , there is locally a scalar c with $ch \in GL_2(R[[\varpi]])$. Since $\mathrm{tr}(h) = 2$ and $2 \in R^\times$, one has $c = \mathrm{tr}(ch)/2 \in R[[\varpi]]$. Moreover $\det(h) = 1 - a\varpi^{2m}$ is a unit, so $\det(ch) = c^2 \det(h)$ being a unit forces $c \in R[[\varpi]]^\times$. Hence h itself is integral, which is exactly $\tilde{\gamma}_m L = L$. Now $\tilde{\gamma}_m - 1 = \varpi^m \iota(z)$, and

$$\tilde{\gamma}_m^{-1} = (1 - a\varpi^{2m})^{-1}(1 - \varpi^m \iota(z)).$$

Therefore preservation by $\tilde{\gamma}_m$ is equivalent to $\mathcal{A}_{m,R}$ -stability. The construction descends because it was phrased in terms of the projective module and its endomorphism.

For (iii), multiplication by u gives the displayed module lift. For a field-valued affine-Grassmannian point, the corresponding PGL_2 -torsor on $\mathcal{O}_F$ is trivial over F . Its Brauer class is therefore in the kernel of $\mathrm{Br}(\mathcal{O}_F) \rightarrow \mathrm{Br}(F)$, which is zero for the regular discrete valuation ring $\mathcal{O}_F$. The remaining vector-bundle lift is free because $\mathcal{O}_F$ is local. After choosing that trivialization, Hilbert 90 applied to $GL_2(F) \rightarrow PGL_2(F) \rightarrow H^1(F, \mathbb{G}_m)$ gives a matrix representative. This proves the two stated lifting cases. $\square$

For $r \geq 0$ put

$$\mathcal{A}_{r,R} = R[[\varpi]] + \varpi^r(\mathcal{E} \otimes_{\mathcal{O}_U} R)[[\varpi]]. \quad (22)$$

This is a rank-two $R[[\varpi]]$ -lattice and a quadratic order. Let $L^+\mathbb{T}$ denote the fppf positive-loop sheaf

$$L^+\mathbb{T}(R) = \frac{(\mathcal{E} \otimes R)[[\varpi]]^\times}{R[[\varpi]]^\times}, \quad (23)$$

and let $L^+\mathbb{T}_r$ be the stabilizer sheaf of $[\mathcal{A}_r]$.

Remark 5.2 (Meaning of “compact shell”). The adjective *compact* refers to the orbit of the maximal compact subgroup $T(\mathcal{O})$, or equivalently to the positive-loop orbit. It does not mean that the representing algebraic U -scheme is proper: after the quadratic splitting cover, $\mathcal{Q}_r$ contains a $\mathbb{G}_m$ factor.

Lemma 5.3 (Local quadratic-order classification). *Let F be a complete discretely valued field of residue characteristic different from two, and let E/F be a split or unramified quadratic étale algebra with maximal order $\mathcal{O}_E$. Put $A_m = \mathcal{O}_F + \varpi^m \mathcal{O}_E$. Every A_m -stable full $\mathcal{O}_F$ -lattice in E , modulo multiplication by $E^\times$, has a unique representative*

$$A_r = \mathcal{O}_F + \varpi^r \mathcal{O}_E, \quad 0 \leq r \leq m.$$

Its multiplier ring is A_r .

Proof. Every fractional $\mathcal{O}_E$ -ideal is principal. Scale a lattice L so that $\mathcal{O}_E L = \mathcal{O}_E$ and hence $L \subset \mathcal{O}_E$. In the field case L contains a unit. In the split case the two projections of $L/\varpi L$ are nonzero; a vector space over a field cannot be the union of its two coordinate-axis intersections, so L again contains an element that is a unit in both factors. Scaling by that unit puts 1 in L .

Choose ω with $\mathcal{O}_E = \mathcal{O}_F \oplus \mathcal{O}_F \omega$. The image of L in $\mathcal{O}_E/\mathcal{O}_F$ is $\varpi^r \mathcal{O}_F \bar{\omega}$ for a unique $r \geq 0$. Subtracting a scalar from a lift of its generator gives $L = \mathcal{O}_F + \varpi^r \mathcal{O}_E = A_r$. Stability under

A_m is equivalent to $\varpi^m \mathcal{O}_E \subset A_r$, hence to $r \leq m$. Finally $x A_r \subset A_r$ implies $x = x \cdot 1 \in A_r$, and the converse holds because A_r is a ring. Thus the multiplier ring is A_r and determines r . $\square$

Lemma 5.4 (The positive-loop orbit is the residual quotient). *For $r \geq 1$ there are canonical isomorphisms of fppf sheaves over U*

$$L^+ \mathbb{T}_r \simeq \mathcal{A}_r^\times / \mathcal{O}_U[[\varpi]]^\times, \quad (24)$$

$$L^+ \mathbb{T} / L^+ \mathbb{T}_r \simeq \frac{\mathcal{E}[[\varpi]]^\times}{\mathcal{O}_U[[\varpi]]^\times (1 + \varpi^r \mathcal{E}[[\varpi]])} \simeq \mathcal{Q}_r. \quad (25)$$

For $r = 0$ the quotient is the point $U = \mathcal{Q}_0$. For every $m \geq r$, the induced orbit morphism

$$L^+ \mathbb{T} / L^+ \mathbb{T}_r \longrightarrow X_m$$

is a monomorphism of fppf sheaves.

Proof. For a U -algebra R , an element of $L^+ \mathbb{T}(R)$ is fppf locally represented by $u \in (\mathcal{E} \otimes R)[[\varpi]]^\times$. It fixes the homothety class of $\mathcal{A}_{r,R}$ precisely when there is $c \in R((\varpi))^\times$ with $u \mathcal{A}_{r,R} = c \mathcal{A}_{r,R}$. Set $a = c^{-1}u$. The displayed equality says $a \mathcal{A}_{r,R} = \mathcal{A}_{r,R}$; since $1 \in \mathcal{A}_{r,R}$, applying the same argument to a^{-1} gives $a \in \mathcal{A}_{r,R}^\times$. Hence $c = u a^{-1}$ belongs to $(\mathcal{E} \otimes R)[[\varpi]]^\times$. Because $\mathcal{E} \otimes R$ is finite faithfully flat over R , scalar Laurent series satisfy

$$R((\varpi))^\times \cap (\mathcal{E} \otimes R)[[\varpi]]^\times = R[[\varpi]]^\times;$$

this is the equalizer form of faithfully flat descent, applied also to the inverse. Thus $c \in R[[\varpi]]^\times$ fppf locally, and the homothety may be absorbed by the scalar subgroup. Conversely every unit of the order preserves it. This proves (24) as an equality of fppf stabilizer sheaves, including the scalar quotient. For every U -algebra R ,

$$\mathcal{A}_{r,R}^\times = R[[\varpi]]^\times (1 + \varpi^r (\mathcal{E} \otimes R)[[\varpi]]).$$

Indeed the reduction of a unit of $\mathcal{A}_{r,R}$ modulo $\varpi^r \mathcal{E}[[\varpi]]$ is a scalar unit, and division by a lift of that scalar leaves an element of the displayed congruence subgroup. Reduction modulo ϖ^r therefore identifies the middle quotient in (25) with

$$\frac{((\mathcal{E} \otimes R)[\varpi]/(\varpi^r))^\times}{(R[\varpi]/(\varpi^r))^\times},$$

which is exactly (12) after $\epsilon \mapsto \varpi$. Every construction commutes with fppf base change, so the calculation for arbitrary R is an isomorphism after fppf sheafification. Finally, if two sections u_1, u_2 have the same image in X_m , then fppf locally $u_2^{-1} u_1$ fixes $[\mathcal{A}_r]$ and hence belongs to $L^+ \mathbb{T}_r$ by the stabilizer calculation. They therefore define the same quotient section. This proves that the orbit map is a monomorphism of fppf sheaves, not only an injection on field-valued points. $\square$

Theorem 5.5 (Represented positive-loop orbit-shell comparison for the universal PGL_2 family). *For $m \geq 0$ and $0 \leq r \leq m$, the order lattice $\mathcal{A}_r$ is fixed by γ_m . The orbit morphism*

$$\theta_{m,r} : \mathcal{Q}_r \xrightarrow{\sim} L^+ \mathbb{T} \cdot [\mathcal{A}_r] \longrightarrow X_m, \quad [u] \longmapsto [u \mathcal{A}_r], \quad (26)$$

identifies $\mathcal{Q}_r$ with the fppf positive-loop orbit sheaf. In particular that orbit sheaf is represented by a smooth finite-type U -scheme, and the induced morphism of fpqc sheaves to $X_m \subset \mathrm{Gr}_G$ is a monomorphism. No representability statement for all of X_m is used here.

Write $\mathcal{O}_{m,r}$ for this represented orbit and set

$$\mathcal{S}_m^{\text{orb}} = \coprod_{r=0}^m \mathcal{O}_{m,r}. \quad (27)$$

By definition of this disjoint union, the componentwise orbit isomorphisms assemble into

$$\Theta_m : U \amalg \coprod_{r=1}^m \mathcal{Q}_r \xrightarrow{\sim} \mathcal{S}_m^{\text{orb}}. \quad (28)$$

This displayed map is a convenient bookkeeping isomorphism, not an additional decomposition theorem for X_m or $[T \backslash X_m]$. For every field-valued point $x \rightarrow U$, with $F_x = k(x)((\varpi))$ and E_x/F_x the induced split or quadratic unramified algebra, the double quotient

$$T_x(F_x) \backslash X_{m,x}(k(x))$$

has exactly the representatives $[A_r]$, $0 \leq r \leq m$. Thus $\mathcal{O}_{m,r}$ is a compact fundamental shell for the corresponding orbital double coset. The theorem does not assert that $[T \backslash X_m]$ is the disjoint union of the $\mathcal{Q}_r$.

Proof. Since $\iota(z)\mathcal{E} = \mathcal{E}$,

$$\varpi^m \iota(z) \mathcal{A}_r \subset \mathcal{A}_r \iff m \geq r.$$

The inverse of $1 + \varpi^m \iota(z)$ is $(1 - a\varpi^{2m})^{-1}(1 - \varpi^m \iota(z))$, so containment is equality. Hence $[A_r] \in X_m$ and the centralizer action preserves X_m . The stabilizer calculation in Lemma 5.4 proves that the fppf orbit is $\mathcal{Q}_r$ and that its orbit map is a monomorphism. Theorem 4.1 supplies smooth finite-type representability. The componentwise isomorphisms assemble tautologically to (28); multiplier rings distinguish the field-valued double-coset representatives indexed by different r .

Over a field-valued point, a coset in X_m is a homothety class of $A_m = \mathcal{O}_{F_x}[\tilde{\gamma}_m]$ -stable lattices. Lemma 5.3 gives precisely the representatives A_r , and their stabilizers are $A_r^\times / \mathcal{O}_{F_x}^\times$. In the anisotropic case $T_x(F_x) = T_x(\mathcal{O}_{F_x})$; in the split case translations along the apartment are supplied by $T_x(F_x)/T_x(\mathcal{O}_{F_x})$. In both cases the compact L^+T -orbit is a fundamental shell whose cardinality represents the inverse stabilizer volume of the double coset. This order calculation is the equal-characteristic form of the rank-one tube/ball decomposition familiar from the explicit formulas in [12, §5.8, (5.8.4); §5.9, (5.9.7)–(5.9.8)]. $\square$

Lemma 5.6 (Full torus stabilizer and orbital support). *Let $x \rightarrow U$ be field-valued, put $F = F_x$, $K = \text{PGL}_2(\mathcal{O}_F)$, and choose g_r with $g_r \mathcal{O}_F^2 = A_{r,x}$. Then*

$$\text{Stab}_{T_x(F)}([A_{r,x}]) = T_x(F) \cap g_r K g_r^{-1} = A_{r,x}^\times / \mathcal{O}_F^\times. \quad (29)$$

Moreover the support of the spherical orbital integrand is the disjoint union

$$\{g \in \text{PGL}_2(F) : g^{-1} \gamma_{m,x} g \in K\} = \coprod_{r=0}^m T_x(F) g_r K. \quad (30)$$

Proof. Write $T_x(F) = E_x^\times / F^\times$. If the class of $e \in E_x^\times$ stabilizes $[A_{r,x}]$, then $eA_{r,x} = cA_{r,x}$ for some $c \in F^\times$. Hence $c^{-1}e$ is a unit of the multiplier ring of $A_{r,x}$, which is $A_{r,x}$ by Lemma 5.3. Conversely every such unit stabilizes the lattice. The kernel of $A_{r,x}^\times \rightarrow E_x^\times / F^\times$ is $A_{r,x}^\times \cap F^\times = \mathcal{O}_F^\times$, proving (29). Finally the set on the left of (30), modulo right translation by K , is $X_{m,x}$. The field-valued double-quotient classification in Theorem 5.5 gives exactly the classes $[A_{r,x}]$; lifting those classes by the chosen g_r proves the disjoint support decomposition. $\square$

Remark 5.7 (Orbit shells are not closure strata of the whole fiber). The schemes $\mathcal{O}_{m,r}$ are compact fundamental representatives for orbital double cosets, not a claimed stratification of all of X_m . This distinction is already visible after the quadratic splitting cover: the closure of the $\mathbb{G}_m$ direction in an ambient Schubert variety acquires boundary points translated along the split apartment, whereas $\mathcal{S}_m^{\text{orb}}$ retains one compact fundamental shell per $T(F)$ -double coset. Consequently neither closure relations nor transition maps may be read off from the disjoint union (27).

Corollary 5.8 (Geometric realization of the spherical orbital integral). *Let $x \in U$ be a closed point with finite residue field k_x , put $F_x = k_x((\varpi))$, and normalize Haar measures by*

$$\text{vol}(PGL_2(k_x[[\varpi]])) = 1, \quad \text{vol}(T_x(k_x[[\varpi]])) = 1.$$

The quotient measure $d\dot{g}$ is the one characterized by Weil integration: for every compactly supported locally constant f ,

$$\int_{PGL_2(F_x)} f(g) dg = \int_{T_x(F_x) \backslash PGL_2(F_x)} \int_{T_x(F_x)} f(tg) dt d\dot{g}. \quad (31)$$

Let Frob_x denote geometric Frobenius. In the last line below, Tr denotes the derived supertrace $\sum_i (-1)^i \text{Tr}(\text{Frob}_x | \mathcal{H}^i)$. Then

$$\begin{aligned} \text{OI}_m(x) &:= \int_{T_x(F_x) \backslash PGL_2(F_x)} \mathbf{1}_{PGL_2(k_x[[\varpi]])} (g^{-1} \gamma_{m,x} g) d\dot{g} \\ &= \sum_{r=0}^m [T_x(k_x[[\varpi]]) : A_{r,x}^\times / k_x[[\varpi]]^\times] = \# \mathcal{S}_{m,x}^{\text{orb}}(k_x) \\ &= \text{Tr} \left(\text{Frob}_x \mid (Rq_{m,!}^{\text{orb}} \mathbf{Q}_\ell)_{\bar{x}} \right), \end{aligned} \quad (32)$$

where $q_m^{\text{orb}} : \mathcal{S}_m^{\text{orb}} \rightarrow U$ is the structure map.

Proof. Put $K_x = PGL_2(k_x[[\varpi]])$ and choose g_r with $g_r k_x[[\varpi]]^2 = A_{r,x}$. By (30), the support of the characteristic function is precisely the disjoint union $\coprod_{r=0}^m T_x(F_x) g_r K_x$. Formula (31) therefore gives

$$\text{vol}(T_x(F_x) \backslash T_x(F_x) g_r K_x) = \frac{\text{vol}(K_x)}{\text{vol}(T_x(F_x) \cap g_r K_x g_r^{-1})}. \quad (33)$$

The full-torus stabilizer calculation (29), not merely the positive-loop stabilizer calculation, identifies the denominator with $A_{r,x}^\times / \mathcal{O}_{F_x}^\times$, as a subgroup of

$$T_x(\mathcal{O}_{F_x}) = \mathcal{O}_{E_x}^\times / \mathcal{O}_{F_x}^\times.$$

Since the latter has volume one, (33) is exactly

$$[T_x(\mathcal{O}_{F_x}) : A_{r,x}^\times / \mathcal{O}_{F_x}^\times].$$

In the unramified-field case $T_x(F_x) = T_x(\mathcal{O}_{F_x})$. In the split case $T_x(F_x)/T_x(\mathcal{O}_{F_x})$ is the apartment-translation lattice; it is already divided out on the left of the quotient, while the stabilizer in (33) remains the same compact subgroup. Thus no additional split-torus volume factor occurs.

Finally, the fiber of $\mathcal{Q}_r$ is the fppf quotient of the unit groups of the two finite k_x -algebras $E_x[\epsilon]/(\epsilon^r)$ and $k_x[\epsilon]/(\epsilon^r)$. The obstruction to replacing quotient-sheaf k_x -points by the naive unit quotient lies in $H^1(k_x, (k_x[\epsilon]/(\epsilon^r))^\times)$, which is the Picard group of this finite local Artin algebra and is zero. Equivalently this is the truncated-unit form of Hilbert 90. Hence $\mathcal{Q}_r(k_x)$ is the ordinary unit quotient and its cardinality is the same index. Sum over the representatives from Theorem 5.5 and apply the Grothendieck–Lefschetz trace formula. $\square$

Proposition 5.9 (Independent Bruhat–Tits verification of the shell support). *At a field-valued point $x \rightarrow U$, let $\mathcal{B}_x$ be the reduced Bruhat–Tits tree of $PGL_2(F_x)$. If T_x is split, let $\mathcal{A}_x$ be its apartment; if T_x is unramified anisotropic, let v_x be its fixed vertex. Then the fixed set of $\gamma_{m,x}$ is, respectively, the closed radius- m tube around $\mathcal{A}_x$ or the closed radius- m ball around v_x . Modulo $T_x(F_x)$ it has one shell at each distance $0 \leq r \leq m$, and the stabilizer of the distance- r shell is $A_{r,x}^\times / \mathcal{O}_{F_x}^\times$.*

Consequently the building calculation independently recovers both the support decomposition (30) and the termwise masses in (32).

Proof. After the unramified splitting extension, the two eigenvalues of $\gamma_{m,x}$ are $1 \pm \varpi^m \sqrt{a}$. Their difference has valuation m . In a lattice chart transverse to the eigenline apartment, moving one edge away divides one off-diagonal integrality condition by ϖ ; hence exactly the vertices of transverse distance at most m are fixed. Galois descent gives the tube in the split case and the ball about the unique anisotropic fixed vertex in the nonsplit case. Apartment translations are already supplied by $T_x(F_x)$, so there is one orbit for each distance.

The lattice at distance r may be represented by $A_{r,x} = \mathcal{O}_{F_x} + \varpi^r \mathcal{O}_{E_x}$. Its multiplier ring is $A_{r,x}$, and the same calculation as in Lemma 5.6 identifies its torus stabilizer with $A_{r,x}^\times / \mathcal{O}_{F_x}^\times$. Weil’s quotient formula then gives the stated masses. This proof uses the metric geometry of the tree rather than the fppf orbit representation, so it is a consistency check independent of the construction of $\mathcal{Q}_r$. $\square$

Theorem 5.10 (Universal PGL_2 compact-orbital reciprocity). *Let $\mathcal{I}_m^{\text{orb}} = Rq_{m,!}^{\text{orb}} \mathbf{Q}_\ell$ and let $I_m^{\text{orb}} = [\mathcal{I}_m^{\text{orb}}]$ be the raw compact-orbital coefficient. Define its Weyl-normalized coefficient by $J_m^{\text{orb}} = Q^m I_m^{\text{orb}}$. The group Weyl discriminant of (19) is as follows; in the absolute value statement x is a finite-field point and $q_x = |k_x|$:*

$$D^G(\gamma_m) = \frac{-4a\varpi^{2m}}{1 - a\varpi^{2m}}, \quad |D^G(\gamma_{m,x})|_x^{1/2} = q_x^{-m}, \quad q_x = |k_x|, \quad (34)$$

so the motivic Weyl factor is $\mathbb{L}_{\text{mot}}^{-m}$ and its ℓ -adic realization is Q^m . In the localized Tate–Kummer Grothendieck ring,

$$J_m^{\text{orb}} = 1 + (1 - \kappa_\ell) \sum_{s=1}^m Q^s, \quad (35)$$

$$W_{PGL_2}(z) := \sum_{m \geq 0} J_m^{\text{orb}} z^m = \frac{1 - \kappa_\ell Qz}{(1 - z)(1 - Qz)}, \quad (36)$$

Moreover,

$$\boxed{W_{PGL_2}(z^{-1}) = -\kappa_\ell z d_{U_k}(W_{PGL_2}(z)), \quad \kappa_\ell = [\det(X_*(\mathbb{T}) \otimes \mathbf{Q}_\ell)] = [\mathcal{K}].} \quad (37)$$

At every finite-field point, the geometric-Frobenius derived supertrace of the coefficient in (36) is the Weyl-normalized actual spherical orbital integral in (32).

Proof. The two eigenvalues of γ_m are $1 \pm \varpi^m \sqrt{a}$, which gives (34); its denominator is a unit on U . This normalization is also the intrinsic root-space Jacobian, not a factor chosen to make the series reciprocal. Indeed, in the standard basis H, E, F of $\mathfrak{pgl}_2 \simeq \mathfrak{sl}_2$, the torus direction is $X_a = aE + F$ and a transverse direction is $Y_a = aE - F$; one has $\det(X_a, H, Y_a) = 2a \in \mathcal{O}_U^\times$. Thus the local slice contributes no additional valuation, while the determinant of the orbit differential on the two root spaces is $D^G(\gamma_m)$.

The componentwise represented-orbit isomorphisms assembled in (28) identify $\mathcal{I}_m^{\text{orb}}$ with the direct sum of the residual complexes. Theorem 4.1 and the calculation in Section 6 give (35)–(37). Corollary 5.8 gives the final trace statement. $\square$

6 The PGL_2 compact-shell generating class

Let $U = U_k = \operatorname{Spec} k[a, a^{-1}, (1-a)^{-1}]$ be the realization base of the quadratic tower and of the PGL_2 family in Sections 4 and 5, and put

$$K_{U_k} = K_0(D_c^b(U_k, \mathbf{Q}_\ell))[\mathbb{L}_\ell^{-1}], \quad \mathbb{L}_\ell = [\mathbf{Q}_\ell(-1)[-2]], \quad Q = \mathbb{L}_\ell^{-1}.$$

For a complex M write $[M]$ for its class; for a finite-type map $f : Y \rightarrow U$ abbreviate $[Y] = [Rf_! \mathbf{Q}_\ell]$. Let $\tilde{U} = \operatorname{Spec} k[b, b^{-1}, (1-b^2)^{-1}] \rightarrow U$, $b^2 = a$, be the quadratic cover. Define $\mathcal{K}$ by $R\tilde{p}_* \mathbf{Q}_\ell \simeq \mathbf{Q}_\ell \oplus \mathcal{K}$, with the deck involution acting by -1 on $\mathcal{K}$, and put $\kappa_\ell = [\mathcal{K}]$. Raw Verdier duality on the smooth curve U is

$$\mathbb{D}_U^{\text{raw}}(M) = R\mathcal{H}om_U(M, \mathbf{Q}_\ell(1)[2]).$$

Define the dimension-normalized coefficient duality by

$$\mathbf{d}_{U_k}(M) := \mathbb{D}_U^{\text{raw}}(M) \otimes^L \mathbf{Q}_\ell(-1)[-2], \quad d_{U_k}([M]) := [\mathbf{d}_{U_k}(M)]. \quad (38)$$

Thus $\mathbf{d}_{U_k}$ fixes the constant coefficient, unlike raw Verdier duality. On the lisse Tate–Kummer subcategory it induces an involution with

$$d_{U_k}(Q) = Q^{-1}, \quad d_{U_k}(\kappa_\ell) = \kappa_\ell, \quad \kappa_\ell^2 = 1. \quad (39)$$

We extend d_{U_k} coefficientwise to rational functions and let it fix the grading variable z ; inversion of z is written separately. The finite-type shell of radius r has the class equality

$$[Rq_{r,!} \mathbf{Q}_\ell] = [R\pi_{\mathbb{T},!} \mathbf{Q}_\ell] \mathbb{L}_\ell^{r-1} = (\mathbb{L}_\ell - \kappa_\ell) \mathbb{L}_\ell^{r-1}$$

in K_{U_k} ; no product isomorphism $\mathcal{Q}_r \simeq \mathbb{T} \times \mathbb{A}^{r-1}$ is asserted. The two terms of $\mathbb{L}_\ell - \kappa_\ell$ are not a formal subtraction invented after counting. They are the two adjacent compactly supported cohomology groups of the one-dimensional torus: the top Tate line and the degree-one orientation line.

Assign the Weyl factor Q^m and put $t = m - r$. The corresponding residual layer contributes

$$Q^m[\mathcal{Q}_r] = Q^{m-r}(1 - \kappa_\ell Q) = Q^t - \kappa_\ell Q^{t+1}. \quad (40)$$

Define the compact-orbital coefficient $I_m^{\text{orb}} = 1 + \sum_{r=1}^m [\mathcal{Q}_r]$; by Theorem 5.5 this is the compact-shell realization of the spherical orbital integral, and the first term is the core. Summing the layers gives

$$\begin{aligned} J_m^{\text{orb}} &:= Q^m I_m^{\text{orb}} = Q^m + \sum_{t=0}^{m-1} (Q^t - \kappa_\ell Q^{t+1}) \\ &= 1 + (1 - \kappa_\ell) \sum_{s=1}^m Q^s. \end{aligned} \quad (41)$$

The coordinate

$$\boxed{s = t + 1 = m - r + 1} \quad (42)$$

is therefore forced by the adjacent cohomological degrees in (40) and by telescoping of the shell filtration. It is not a relabeling chosen to make a denominator palindromic; no analogous formula for a general orbital family is inferred from this calculation.

Set $A = 1 - \kappa_\ell$ and let

$$C = \{(m, s) : m \geq 0, 0 \leq s \leq m\}, \quad F = \{(m, 0) : m \geq 0\}.$$

For an explicitly height-graded finite array whose (m, s) component has been written as $V_{m,s}(s)$, define its de-Tated height character by

$$\text{ch}_{\text{ht}}(V) := \sum_{m,s} [V_{m,s}] z^m u^s. \quad (43)$$

This operation is defined on the supplied height grading; it is not a functor on the ungraded derived category and does not identify the twist (s) with a constant coefficient object. Specialization $u \mapsto Q$ recovers the ordinary Grothendieck class because $[V(s)] = Q^s[V]$ in the present convention.

Here $\mathbf{1}_C$ and $\mathbf{1}_F$ denote the constant closed-cone cellular generators on the finite rational fan with maximal cone C and face F ; they are not direct sums of one object for every lattice point. Define the auxiliary untwisted cellular envelope

$$\hat{\Phi} = A \mathbf{1}_C + \kappa_\ell \mathbf{1}_F \in K_0 \text{Cell}_C(D_c^b(U, \mathbf{Q}_\ell)). \quad (44)$$

It encodes the de-Tated height character extracted from (41); it is not asserted to be the ordinary cellularization of the Tate-twisted shell complexes. Its infinite lattice expansion is interpreted only through the rational generating transform in the localization generated by $1 - z$ and $1 - uz$. Thus no infinite coproduct in a bounded cellular category is being asserted. The bivariate lattice transform of $\hat{\Phi}$ and its cellular partner are

$$\mathcal{W}(z, u) = \frac{A}{(1-z)(1-uz)} + \frac{\kappa_\ell}{1-z} = \frac{1 - \kappa_\ell uz}{(1-z)(1-uz)}, \quad (45)$$

$$\mathcal{W}^!(z, u) = A \frac{uz^2}{(1-z)(1-uz)} - \kappa_\ell \frac{z}{1-z}. \quad (46)$$

Theorem A.1, with the displayed integral orientations, gives

$$\mathcal{W}^!(z^{-1}, u^{-1}) = \mathcal{W}(z, u). \quad (47)$$

Direct simplification, using $-\kappa_\ell(1 - \kappa_\ell) = 1 - \kappa_\ell$, also gives the special self-partner identity

$$\mathcal{W}^!(z, u) = -\kappa_\ell z \mathcal{W}(z, u). \quad (48)$$

Specialize $u \mapsto Q$ and write $W_{PGL_2}(z) = \mathcal{W}(z, Q)$. Apply d_{U_k} to the two preceding identities, using (39). This yields the unambiguous one-variable formula

$$\boxed{W_{PGL_2}(z^{-1}) = -\kappa_\ell z d_{U_k}(W_{PGL_2}(z)), \quad \kappa_\ell = [\mathfrak{o}_{\mathbb{T}}] = [\mathcal{K}], \quad \mathfrak{o}_{\mathbb{T}} = \det(X_*(\mathbb{T}) \otimes \mathbf{Q}_\ell).} \quad (49)$$

Equation (49) is an unconditional rational K_0 identity assembled from finite-type schemes. Theorem 5.5 identifies those schemes with the compact positive-loop orbit shells that represent the actual spherical PGL_2 double-coset masses. Thus the equation is an orbital reciprocity theorem for this family. It is not a theorem about the entire derived quotient stack of the affine-Springer locus.

Proposition 6.1 (Interpretation of the three factors). *In (49):*

- (i) d_{U_k} is the coefficient involution induced by dimension-normalized Verdier duality;
- (ii) $\mathfrak{o}_{\mathbb{T}}$ is the determinant of the norm-one torus cocharacter local system and records quadratic descent;
- (iii) $-z$ is the localized- K_0 shadow of the canonical module of the depth ray, including its odd shift and primitive translation.

The coordinate $s = m - r + 1$ belongs to the preceding orbital-shell-to-skeleton bookkeeping step and is not any one of these three factors.

Proof. Part (i) is the definition in (39). Part (ii) is the quadratic sign action on $X_*(\mathbb{T})$, together with Theorem 3.2; it uses no orbital product chart. Part (iii) is the canonical module of $\mathbf{N}$. Equation (42) was derived before applying polyhedral duality, so it belongs to the orbital shell grading convention, not to the same-series identification (48). $\square$

Warning 6.2 (The functional equation is still a K_0 statement). The class $A = 1 - \kappa_\ell$ is virtual, and the denominators in (45) represent an infinite grading. The equality $-\kappa_\ell(1 - \kappa_\ell) = 1 - \kappa_\ell$ uses cohomological signs in K_0 and is not an isomorphism between the naive bounded complexes representing the two sides. The finite labelled-shell theorem below gives an object-level partner equivalence on its discrete target. The more polyhedral height object exists only after cohomological associated graded, and its assembled horizontal duality is asserted only after K_0 . Neither theorem constructs cross-shell transition maps or an incidence-bearing ungraded equivalence. Completion is a separate operation even for an eventually constant finite filtration: t -adic completion generally destroys the $t = 1$ specialization. An infinite filtration adds further continuity and convergence questions.

7 A finite labelled-shell Verdier partner and its polyhedral K_0 shadow

This section separates two statements of different strength. First, the actual finite family of labelled orbit shells admits a genuine zero-dimensional Verdier partner: both direct-image functors are defined independently and Verdier duality intertwines them by a natural equivalence. Second, the height regrading produces a more polyhedral object only after cohomological associated graded; its assembled horizontal duality remains a K_0 statement.

7.1 The finite labelled-shell Verdier partner

Fix $N \geq 0$ and let

$$\Lambda_N = \{(m, r) : 0 \leq r \leq m \leq N\} \quad (50)$$

with the discrete finite cell structure, every point being a zero-cell. Put

$$\mathcal{Y}_N^{\text{orb}} = \coprod_{(m,r) \in \Lambda_N} \mathcal{O}_{m,r}, \quad \mathcal{Z}_N = U \times \Lambda_N = \coprod_{(m,r) \in \Lambda_N} U, \quad (51)$$

Write $q_{m,r} : \mathcal{O}_{m,r} \rightarrow U$ for the structure map and define the labelled shell map

$$\rho_N : \mathcal{Y}_N^{\text{orb}} \longrightarrow \mathcal{Z}_N, \quad x \in \mathcal{O}_{m,r} \longmapsto (q_{m,r}(x), (m, r)). \quad (52)$$

This map is constructed from the actual orbit schemes and their structure maps; the target is not inferred from a generating function.

There is a canonical equivalence

$$D_c^b(\mathcal{Z}_N, \mathbf{Q}_\ell) \simeq \text{Cell}_{\Lambda_N}(D_c^b(U, \mathbf{Q}_\ell)) \simeq \prod_{(m,r) \in \Lambda_N} D_c^b(U, \mathbf{Q}_\ell). \quad (53)$$

Let $\mathbb{D}_{\text{cell},N}$ denote total cellular duality on this category. Because all cells have dimension zero, it is componentwise Verdier duality $\mathbb{D}_U$; there is no additional cellular shift or orientation line. Define the two functors independently by

$$\text{Sk}_{N,!} = R\rho_{N,!}, \quad \text{Sk}_{N,*} = R\rho_{N,*} : D_c^b(\mathcal{Y}_N^{\text{orb}}, \mathbf{Q}_\ell) \longrightarrow \text{Cell}_{\Lambda_N}(D_c^b(U, \mathbf{Q}_\ell)), \quad (54)$$

using (53). In particular $\text{Sk}_{N,*}$ is not defined as a Verdier conjugate of $\text{Sk}_{N,!}$.

Theorem 7.1 (Finite labelled-shell Verdier partner). *For every finite cutoff N , the functors in (54) are defined on the full bounded constructible category and there is a canonical natural equivalence of contravariant functors*

$$\boxed{\mathbb{D}_{\text{cell},N} \circ \text{Sk}_{N,!} \simeq \text{Sk}_{N,*} \circ \mathbb{D}_{\mathcal{Y}_N^{\text{orb}}} .} \quad (55)$$

For $M = (M_{m,r})$ under the finite disjoint-union decomposition, its (m,r) component is precisely

$$\mathbb{D}_U Rq_{m,r,!} M_{m,r} \simeq Rq_{m,r,*} \mathbb{D}_{\mathcal{O}_{m,r}} M_{m,r}. \quad (56)$$

The equivalences are compatible with restriction from $N+1$ to N .

Proof. Each $\mathcal{O}_{m,r} \simeq \mathcal{Q}_r$ is a smooth affine finite-type U -scheme by Theorems 4.1 and 5.5. Hence ρ_N is separated and of finite type, and both direct images preserve bounded constructibility. The six-functor Verdier isomorphism for this independently specified map gives

$$\mathbb{D}_{\mathcal{Y}_N} R\rho_{N,!} \simeq R\rho_{N,*} \mathbb{D}_{\mathcal{Y}_N^{\text{orb}}}$$

naturally on the full source category. Under (53), $\mathbb{D}_{\mathcal{Y}_N}$ is exactly $\mathbb{D}_{\text{cell},N}$, proving the boxed equivalence. Decomposition over the zero-cells gives (56). The inclusion $\Lambda_N \subset \Lambda_{N+1}$ is a union of open-and-closed components, so restriction preserves the construction and its Verdier transformation. $\square$

Remark 7.2 (Why the theorem is genuine and why it is small). The source, target, map, two functors, and both dualities are all fixed before the equivalence is invoked, and the equivalence is natural in every constructible coefficient object. Thus Theorem 7.1 is an object-level Verdier partner identity, not a K_0 identity. It is small because Λ_N is discrete: the theorem records the actual labels (m,r) but no closure incidence, no height translation $s = m - r + 1$, and no Stanley canonical-module operation. The split product chart is not needed for this theorem.

7.2 The associated-graded polyhedral shadow

We next retain the canonical height data in addition to the shell labels. Let

$$C = \{(m,s) \in \mathbf{N}^2 : 0 \leq s \leq m\}, \quad F = \{(m,0) : m \geq 0\}.$$

For a complex M , write $\text{gr}_{\mathcal{H}} M = \bigoplus_i \mathcal{H}^i(M)[-i]$. This associated graded is canonical; it does not assert that M splits.

Retaining the structure maps $q_{m,r}$ above, put

$$B_{m,0} = \mathbf{Q}_{\ell}(m), \quad B_{m,r} = Rq_{m,r,!} \mathbf{Q}_{\ell}(m)[2r] \quad (1 \leq r \leq m). \quad (57)$$

The shift only places the two compact-cohomology sheaves in degrees -1 and 0 ; its class is unaffected.

Theorem 7.3 (Associated-graded coefficient array and its K_0 polyhedral shadow). *For the actual compact orbit shells of Theorem 5.5, the following statements hold.*

- (i) *For $r \geq 1$ there is a canonical identification of cohomology sheaves*

$$\text{gr}_{\mathcal{H}} B_{m,r} \simeq \mathbf{Q}_{\ell}(m-r) \oplus \mathcal{K}(m-r+1)[1]. \quad (58)$$

Consequently the canonical translation $s = m - r + 1$ gives

$$\bigoplus_{r=0}^m \text{gr}_{\mathcal{H}} B_{m,r} \simeq \mathbf{Q}_{\ell} \oplus \bigoplus_{s=1}^m (\mathbf{Q}_{\ell} \oplus \mathcal{K}[1])(s). \quad (59)$$

- (ii) For a finite cutoff N , put $C_N = \{(m, s) \in C : m \leq N\}$ and define the discrete graded-array category

$$\mathrm{Gr}_{C_N}(U) := \prod_{(m,s) \in C_N} D_c^b(U, \mathbf{Q}_\ell). \quad (60)$$

The component formula

$$\Psi_{N;m,s}^{\mathrm{gr}} = \begin{cases} \mathbf{Q}_\ell, & s = 0, \\ (\mathbf{Q}_\ell \oplus \mathcal{K}[1])(s), & 1 \leq s \leq m \end{cases} \quad (61)$$

defines an object of (60), and (59) identifies it with the cohomological associated graded of the normalized actual shell coefficients. This is a discrete coefficient array, not yet a cellular sheaf on the incidence-bearing cone C .

- (iii) Apply the de-Tated height character (43) to (61). The resulting class function is the restriction to C_N of the auxiliary untwisted cellular envelope

$$\widehat{\Phi} = (1 - \kappa_\ell) \mathbf{1}_C + \kappa_\ell \mathbf{1}_F \in K_0 \mathrm{Cell}_C(D_c^b(U, \mathbf{Q}_\ell)), \quad (62)$$

from (44). This is not an equality between the Tate-twisted shell array and an ordinary cellular object: the twist is recorded by u^s , and $u \mapsto Q$ recovers its ordinary K_0 class. Nor is $\widehat{\Phi}$ the image of a constructed geometric functor to C . Cellular duality and the lattice transform of this auxiliary envelope give (37) after rational continuation and the specialization $u \mapsto Q$.

Proof. The comparison theorem reduces the compact cohomology to H_r . The two nonzero cohomology sheaves of $R\pi_{\mathbb{T},!} \mathbf{Q}_\ell$ are

$$\mathcal{H}^1 = \mathcal{K}, \quad \mathcal{H}^2 = \mathbf{Q}_\ell(-1).$$

The vector-torsor Thom shift in (14) therefore gives

$$\mathcal{H}^{2r-1}(Rq_{m,r,!} \mathbf{Q}_\ell) = \mathcal{K}(-(r-1)), \quad \mathcal{H}^{2r}(Rq_{m,r,!} \mathbf{Q}_\ell) = \mathbf{Q}_\ell(-r).$$

Twisting by (m) and shifting by $[2r]$ proves (58). Its Tate terms have heights $m-r = 0, \dots, m-1$, the core supplies height m , and its Kummer terms have heights $m-r+1 = 1, \dots, m$. This proves (59) and (61).

After removing the displayed twist (s) into the formal height monomial u^s , the class of $\mathbf{Q}_\ell \oplus \mathcal{K}[1]$ is $1 - \kappa_\ell$, while the class at $s = 0$ is $1 = (1 - \kappa_\ell) + \kappa_\ell$. This proves the height-character statement (62); it would be false as a literal identification of the Tate-twisted array with an untwisted ordinary cellular object. The calculation in Section 6, equivalently Theorem A.1 applied to the auxiliary envelope and then specialized by $u \mapsto Q$, gives the functional equation. $\square$

Remark 7.4 (The split product chart computes the vertical partner only). After the Kummer cover $b^2 = a$, the ratio of the two split factors gives an explicit chart

$$\mathcal{Q}_r \times_U \widetilde{U} \simeq \mathrm{Res}_{\widetilde{U}[\varpi]/(\varpi^r)/\widetilde{U}} \mathbb{G}_m \simeq \mathbb{G}_m \times \mathbb{A}^{r-1}. \quad (63)$$

Thus compact and ordinary push-forwards are independently computed by the product chart. Fixed-map Verdier duality $\mathbb{D}_U Rq_! \simeq Rq_* \mathbb{D}$ and Theorem 3.2 identify their vertical coefficient partners. The deck involution interchanges the two quadratic factors; it acts by -1 on the rank-one torus cocharacter and hence descends as $\mathcal{K}$. All product-chart pairings are equivariant, so the shellwise object comparison descends to U . The chart supplies no geometric map to C and no incidence morphisms between distinct (m, r) .

Warning 7.5 (The zero-dimensional theorem does not prove the cone theorem). The source and target of Theorem 7.1 are fixed independently, so its partner equivalence is genuine. By contrast, Theorem 7.3 deliberately passes to cohomological associated graded and then to a cellular envelope only after K_0 . It defines no geometric skeleton map to C , no functor on arbitrary orbital morphisms, and no cross-shell incidence maps. It therefore proves neither an object-level positive-dimensional identity $\mathbb{D}_{\text{cell}} \text{Sk}_! \simeq \text{Sk}_* \mathbb{D}$, nor a Rees action or compatibility with convolution.

A Cellular duality and the generating-function shadow

Let N be a lattice and let Σ be a finite pointed rational fan in $N_{\mathbf{R}}$. For a cell σ , let $\mathbf{k}_{\bar{\sigma}}$ denote the constant sheaf on its closure, extended by zero, and let $\mathbf{k}_{\sigma^\circ}$ denote extension by zero from its relative interior. Scalar cellular duality gives

$$\mathbb{D}_{\text{cell}} \mathbf{k}_{\bar{\sigma}} \simeq \text{or}_\sigma[\dim \sigma] \otimes \mathbf{k}_{\sigma^\circ}. \quad (64)$$

The link complex, rather than a rank-one interior indicator, is used at a singular face. Cellular sheaf–cosheaf duality and subdivision invariance are developed in [8]; see [9] for a correction to the poset Kan-extension argument. Duality and direct image for sheaves on fans are also treated in [1].

Let $\mathcal{C}$ be a coefficient category with a duality d . We use $\text{Cell}_\Sigma(\mathcal{C})$ for $\mathcal{C}$ -valued cellular diagrams and write $\mathbb{D}_{\mathcal{C}, \Sigma}^{\text{tot}} = d \boxtimes \mathbb{D}_{\text{cell}}$. Thus

$$\mathbb{D}_{\mathcal{C}, \Sigma}^{\text{tot}}(V \boxtimes \mathbf{k}_{\bar{\sigma}}) \simeq d(V) \otimes \text{or}_\sigma[\dim \sigma] \otimes \mathbf{k}_{\sigma^\circ}.$$

The orientation line in (64) is part of $\mathbb{D}_{\text{cell}}$; it must not be silently replaced by a scalar sign when the integral lattice has monodromy. Define the coefficient–orientation involution on closed-cell generators by

$$\mathfrak{d}_\Sigma^{\text{or}}(V \otimes \mathbf{k}_{\bar{\sigma}}) = d(V) \otimes \text{or}_\sigma \otimes \mathbf{k}_{\bar{\sigma}}. \quad (65)$$

Unlike $\mathbb{D}_{\mathcal{C}, \Sigma}^{\text{tot}}$, this operation does not exchange a closed cell with its interior.

For $\Phi \in K_0 \text{Cell}_\Sigma(\mathcal{C})$, its stalk class is a cellwise-constant function

$$\varphi_\Phi : N \cap |\Sigma| \longrightarrow K_0(\mathcal{C}).$$

When the resulting series is rational, define

$$\text{GF}(\Phi)(z) = \sum_{u \in N \cap |\Sigma|} \varphi_\Phi(u) z^u \quad (66)$$

in the localization generated by the relevant factors $1 - z^v$.

Theorem A.1 (Generating functions transport cellular duality). *For every finite K_0 -combination of closed rational cones with coefficients in $\mathcal{C}$,*

$$\boxed{\text{GF}(\mathbb{D}_{\mathcal{C}, \Sigma}^{\text{tot}} \Phi)(z^{-1}) = \text{GF}(\mathfrak{d}_\Sigma^{\text{or}} \Phi)(z)}. \quad (67)$$

The equality is independent of rational polyhedral subdivision. If the integral orientations are compatibly trivialized, the right side reduces to $d(\text{GF}(\Phi)(z))$.

Proof. The classes $[V \otimes \mathbf{k}_{\bar{\sigma}}]$ generate the cellular Grothendieck group. Put $e = \dim \sigma$. On this generator the left side is

$$(-1)^e [d(V) \otimes \text{or}_\sigma] \sum_{u \in \text{relint}(\sigma) \cap N} z^{-u}.$$

Rational cone reciprocity [17] gives

$$\sum_{u \in \text{relint}(\sigma) \cap N} z^{-u} = (-1)^e \sum_{u \in \sigma \cap N} z^u.$$

The two signs cancel, proving (67) on generators and hence by additivity in general. Under a subdivision, inclusion–exclusion on closed cells and its Verdier dual include all internal faces with the correct link multiplicity. Equivalently, the constructible stalk function and the dualizing complex do not depend on the chosen cellular presentation. $\square$

B The local unramified absolute- A_1 theorem

The represented PGL_2 family separates the Kummer line from the Tate coefficient. We now test whether that line survives changes of central isogeny and shell lattice. This section is local: except in the saturated PGL_2 case, its coefficient objects are explicit orbit-summed interpolation classes, not relative orbital push-forwards.

Let S/k be a smooth connected cameral parameter base of dimension d over a finite field k , with $\ell \neq \text{char } k$. Write $\mathbb{L}_\ell$ also for the class $[\mathbf{Q}_\ell(-1)[-2]]$ over S , and put

$$\mathcal{R}_S = K_0(D_{\text{lis}, \text{Weil}}^b(S, \mathbf{Q}_\ell)) \otimes_{\mathbf{Z}} \mathbf{Q}, \quad Q = \mathbb{L}_\ell^{-1}.$$

Normalize coefficient duality by

$$d_S(M) = \mathbb{D}_S(M)(-d)[-2d]. \quad (68)$$

Thus d_S is contragredience on lisse objects and sends Q to Q^{-1} . All rational identities below lie in the d_S -stable localization

$$\Lambda_S = \mathcal{R}_S[z^{\pm 1}, (1-z)^{-1}, (1+z)^{-1}, (1-Qz)^{-1}, (1-Q^{-1}z)^{-1}]. \quad (69)$$

Let $\tilde{S} \rightarrow S$ be the quadratic cameral cover of the root torus. Its sign line is the rank-one local system $\mathfrak{o}$; at a closed point x write $\chi_x \in \{1, -1\}$ for its geometric-Frobenius trace. The deck involution acts on the root cocharacter line by -1 , so throughout this section

$$\mathfrak{o} \simeq \mathfrak{o}_{G,T} = \det(X_*(T/Z(G)^\circ) \otimes \mathbf{Q}_\ell). \quad (70)$$

Remark B.1 (Central neutrality is necessary). For $G = GL_2$, multiplying a standard lift by ϖI preserves its image in PGL_2 , quadratic order, fixed set, Weyl discriminant, and root orientation. Its determinant then has nonzero valuation, so no conjugate lies in $GL_2(\mathcal{O}_F)$ and the spherical orbital integral vanishes. Condition (2) removes exactly this connected-center counterexample; alternatively one would have to change the test function to one compact modulo center.

Let $p : G \rightarrow G_{\text{ad}}$, put $H = p(G(F))$, and let $\mathcal{B}$ be the reduced Bruhat–Tits tree of $PGL_2(F)$.

Lemma B.2 (Finite central lifting over the hyperspecial model). *Put $G' = G/Z(G)^\circ$ and let $\mathcal{Z}^\circ$ be the smooth connected hyperspecial model of the unramified torus $Z(G)^\circ$. After removing the connected center, the induced central isogeny has the form*

$$1 \longrightarrow Z_{\text{fin}} \longrightarrow G' \longrightarrow G_{\text{ad}} \longrightarrow 1, \quad |Z_{\text{fin}}| \leq 2. \quad (71)$$

For the hyperspecial models one has

$$H^1(\mathcal{O}_F, \mathcal{Z}^\circ) = 1, \quad H^1(\mathcal{O}_F, Z_{\text{fin}}) \hookrightarrow H^1(F, Z_{\text{fin}}).$$

Consequently an element of $G_{\text{ad}}(\mathcal{O}_F)$ which has a lift to $G(F)$ already lifts to $K = \mathcal{G}(\mathcal{O}_F)$.

Proof. The connected center is an unramified torus. Its smooth integral model has connected special fiber, so Lang's theorem followed by henselian lifting gives $H^1(\mathcal{O}_F, \mathcal{Z}^\circ) = 1$. The remaining kernel is a quotient of the center of the simply connected group of type A_1 , hence has order at most two. In odd residue characteristic it is finite étale. A finite étale torsor over the henselian discrete valuation ring which is trivial on the generic fiber is already trivial over the ring, proving the displayed injectivity. The final assertion is the connecting-class criterion applied first to the finite central isogeny and then to the connected center. $\square$

Lemma B.3 (Classification reduction for the reduced-building data). *For every pair and family in Definitions 2.2 and 2.3:*

- (a) $G_{\text{ad}} \simeq PGL_2$, $G_{\text{der}}^{\text{sc}} \simeq SL_2$, and multiplication induces a central isogeny $Z(G)^\circ \times SL_2 \rightarrow G$.
- (b) The type character

$$\tau : PGL_2(F) \longrightarrow \mathbf{Z}/2, \quad [g] \longmapsto \text{ord}_F(\det g) \pmod{2}, \quad (72)$$

has $\tau(H) = 0$ or $\mathbf{Z}/2$. In the first case H preserves vertex color; in the second it sees both colors.

- (c) The root torus T_{rt} is split or the norm-one torus of the unramified quadratic extension. Its orientation line is, respectively, the constant line or the unramified quadratic sign line.
- (d) If $K_{\text{ad}} = G_{\text{ad}}(\mathcal{O}_F)$, then

$$p(K) = H \cap K_{\text{ad}}, \quad T(F) \backslash G(F) \xrightarrow{\sim} (H \cap T_{\text{rt}}(F)) \backslash H. \quad (73)$$

- (e) The shell lattice is saturated when $\tau(H) = \mathbf{Z}/2$ and type-preserving when $\tau(H) = 0$. In the type-preserving elliptic case there are exactly two rational embedding phases, according to the color of the fixed vertex; they retain the even or the odd spheres. This list is exhaustive under the two definitions above.

Proof. The A_1 Dynkin diagram has no nontrivial automorphism. Hence the quasi-split adjoint form is split PGL_2 , its simply connected cover is SL_2 , and the center-derived-group decomposition proves (a) [15, Proposition 20.32 and Theorem 23.51]; see also [6, Appendix C.4].

The character (72) is well defined because scalar multiplication changes determinant valuation by an even integer. The image H contains $PSL_2(F)$, which is transitive on the vertices of either fixed color. This proves (b). Unramified maximal tori in PGL_2 come from the split quadratic algebra or the unramified quadratic field, and Galois acts by $+1$ or -1 on their cocharacter lines, proving (c).

The kernel of $G(F) \rightarrow H$ lies in $T(F)$, which gives the quotient homeomorphism in (d). Lemma B.2 shows that an integral adjoint point with a generic lift already lifts integrally; hence $p(K) = H \cap K_{\text{ad}}$. Finally the split torus fixes an apartment containing both colors, whereas an unramified elliptic torus fixes one vertex. Intersecting the standard tube or ball with the H -orbit retains all spheres, the even spheres, or the odd spheres. A type-changing element exchanges the two elliptic phases, and $PSL_2(F)$ is transitive within each color. This proves (e) and exhaustiveness. $\square$

The preceding lemma classifies only reduced-building data. It does not identify the number or volume of the $T(F)$ -orbits inside a sphere; that is the content of the next lemma.

Lemma B.4 (Adjoint quotient measure and orbit-summed shell transfer). *Put*

$$T_H = H \cap T_{\text{rt}}(F), \quad K_H = H \cap K_{\text{ad}}.$$

Normalize $\text{vol}_G(K) = 1$ and $\text{vol}_T(T_c) = 1$, where $T_c = T(F)_{\text{max,cpt}}$. On H and T_H use the quotient Haar measures along $Z(G)(F)$ with its maximal compact subgroup of volume one. Then $\text{vol}_H(K_H) = 1$ and $\text{vol}_{T_H}(T_{H,c}) = 1$, where $T_{H,c} = T_H \cap T_{\text{rt}}(F)_{\text{max,cpt}} = p(T_c)$. The homeomorphism (73) preserves the resulting quotient measures. If the root torus has Frobenius sign $\chi \in \{1, -1\}$, every allowed positive sphere of radius r has total mass

$$\sum_{\mathcal{O} \text{ over the radius-} r \text{ sphere}} \text{vol}(T_H \cap g\mathcal{O}K_Hg\mathcal{O}^{-1})^{-1} = (q - \chi)q^{r-1}. \quad (74)$$

The radius-zero core, when allowed, has mass one. Every radius is allowed in the saturated case. In a type-preserving elliptic phase ϵ , exactly the radii $r \equiv \epsilon \pmod{2}$ are allowed; in the split case every transverse radius is allowed.

Proof. Put $Z = Z(G)(F)$ and $Z_c = Z \cap K$. The latter is the maximal compact subgroup of Z : the connected center is an unramified torus and the remaining finite central subgroup is integral in odd residue characteristic. Normalize its Haar measure by $\text{vol}_Z(Z_c) = 1$. The tori T and $Z(G)^\circ$ are unramified. Indeed the central-isogeny description and the unramified root torus give unramified character lattices up to finite index. For an unramified torus the valuation map has kernel its maximal compact subgroup and image the Frobenius-fixed cocharacter lattice; $\lambda(\varpi)$ realizes each such valuation.

We verify $p(T_c) = T_{H,c}$. If $p(t)$ is compact, the valuation of t maps to zero in the root cocharacter lattice. It is therefore a Frobenius-fixed cocharacter λ of $Z(G)^\circ$. Multiplication by $\lambda(\varpi)^{-1}$ changes t to an element of T_c without changing its adjoint image. The reverse inclusion follows from compactness. Both $K \rightarrow K_H$ and $T_c \rightarrow T_{H,c}$ are thus surjective with kernel Z_c , the former by Lemma B.2. Weil's formula gives volume one to both image subgroups. Transitivity of quotient integration along $Z \subset T(F) \subset G(F)$ identifies the two quotient measures in (73). In particular this argument makes no use of a radius-zero shell. For any $g \in G(F)$, the map $T(F) \cap gKg^{-1} \rightarrow T_H \cap p(g)K_Hp(g)^{-1}$ is surjective with kernel Z_c : an integral lift of its conjugate lies in $T(F)$ because its adjoint image lies in T_{rt} . Its two stabilizer volumes agree.

In the elliptic case $T_H = T_{H,c}$ is compact of volume one. If Σ_r is an allowed sphere about the fixed vertex, each orbit has mass equal to its cardinality, since its stabilizer is an open subgroup of this volume-one torus. Thus

$$\sum_{\mathcal{O} \subset \Sigma_r} \text{vol}(T_H \cap g\mathcal{O}K_Hg\mathcal{O}^{-1})^{-1} = |\Sigma_r| = (q + 1)q^{r-1} \quad (r \geq 1).$$

When its color is allowed, the fixed vertex has stabilizer T_H and mass one; in the odd phase it is absent. For example, take SL_2 , $q = 5$, and an elliptic fixed vertex adjacent to the vertex of K . In this odd phase the first sphere has two three-point torus orbits. Their stabilizers have volume $1/3$, so their masses sum to $3 + 3 = 6$, although $T(F) \cap K$ itself has volume $1/3$.

In the split case choose coordinates identifying the root torus with the diagonal torus, and index its apartment vertices by $n \in \mathbf{Z}$. This change of coordinates preserves H : it contains $PSL_2(F)$ and is the inverse image of a subgroup of the abelian determinant-square-class quotient, hence is normal in $PGL_2(F)$. The translation image of T_H contains $2\mathbf{Z}$, since H contains $PSL_2(F)$. If H has an element $[g]$ of odd determinant valuation, put $a = \det g$. The elements $[g]$ and $[\text{diag}(a, 1)]$ have the same determinant square class; their quotient belongs to $PSL_2(F)$. Hence the latter diagonal element belongs to H . The translation lattice is consequently $\mathbf{Z}$ in the saturated case and $2\mathbf{Z}$ in the type-preserving case. The kernel of translation is $T_{H,c}$ and fixes the apartment pointwise.

A vertex at transverse distance r with projection n has color $n + r$ modulo two, with the origin chosen to have the hyperspecial color. For a translation period of length one there is one projection vertex; for a period of length two, exactly one of $n = 0, 1$ has the allowed color at each fixed r . Thus in either case a fundamental section of the allowed vertices has one core vertex at $r = 0$ and $(q - 1)q^{r-1}$ vertices at $r \geq 1$. Within this section the remaining orbits are those of $T_{H,c}$. A vertex stabilizer has zero translation, so it lies in $T_{H,c}$; its inverse volume equals its orbit cardinality. Summing gives the asserted transverse mass and core mass one. Finally, Lemma B.3 gives the elliptic parities. $\square$

Combining Lemma B.4 with (2) removes the connected center from the spherical orbital integral. For a closed point $x \rightarrow S$, replace $(\mathbb{L}_\ell, \mathfrak{o})$ by (q_x, χ_x) in the following displayed shell classes. Their Frobenius traces are the corresponding local orbital masses. In the saturated PGL_2 case they are the genuine residual push-forwards of Theorem A; otherwise they are the announced orbit-summed interpolation.

B.1 The SL_2 parity test

The reduced buildings of SL_2 and PGL_2 agree, but $SL_2(F)$ preserves vertex color. This changes the elliptic shell semigroup without changing the root orientation.

Proposition B.5 (Type-preserving shell coefficients). *With $\text{vol}_G(K) = \text{vol}_T(T(F)_{\max, \text{cpt}}) = 1$ in both embedding phases, at a split point,*

$$I_{+,m} = \mathbb{L}_\ell^m, \quad W_+(z) = \frac{1}{1-z}. \quad (75)$$

At an unramified elliptic point let $\epsilon = 0$ when the fixed vertex has the hyperspecial color and $\epsilon = 1$ otherwise. If $m = 2e$ or $2e + 1$, then phase 0 has

$$\begin{aligned} I_{-,2e}^{(0)} = I_{-,2e+1}^{(0)} &= \sum_{r=0}^{2e} \mathbb{L}_\ell^r, \\ J_{-,2e}^{(0)} &= \sum_{s=0}^{2e} Q^s, & J_{-,2e+1}^{(0)} &= \sum_{s=1}^{2e+1} Q^s, \end{aligned} \quad (76)$$

whereas phase 1 has

$$\begin{aligned} I_{-,2e}^{(1)} &= \sum_{r=0}^{2e-1} \mathbb{L}_\ell^r, & J_{-,2e}^{(1)} &= \sum_{s=1}^{2e} Q^s, \\ I_{-,2e+1}^{(1)} &= \sum_{r=0}^{2e+1} \mathbb{L}_\ell^r, & J_{-,2e+1}^{(1)} &= \sum_{s=0}^{2e+1} Q^s. \end{aligned} \quad (77)$$

The empty sum gives $I_{-,0}^{(1)} = 0$.

Proof. In the split case, passing to one vertex color changes translations along the apartment but removes no transverse distance. Lemma B.4 gives total positive shell mass $(\mathbb{L}_\ell - 1)\mathbb{L}_\ell^{r-1}$, and adding the core yields $\mathbb{L}_\ell^r$. In elliptic phase 0 the ball meets the chosen color in the even spheres, including the core; in phase 1 it meets it in the odd spheres and excludes the core. Summing the masses $(\mathbb{L}_\ell + 1)\mathbb{L}_\ell^{r-1}$ over the allowed radii gives the displayed I 's. Multiplication by Q^m gives the J 's. $\square$

Put

$$e_+ = \frac{1 + \mathfrak{o}_{G,T}}{2}, \quad e_- = \frac{1 - \mathfrak{o}_{G,T}}{2}.$$

The two elliptic generating functions and their split/elliptic interpolation are

$$W_-^{(0)}(z) = \frac{1 + Qz^2}{(1 - z^2)(1 - Qz)}, \quad (78)$$

$$W_-^{(1)}(z) = \frac{(1 + Q)z}{(1 - z^2)(1 - Qz)}, \quad (79)$$

$$W_{\text{tp}}^{(\epsilon)}(z) = e_+ W_+(z) + e_- W_-^{(\epsilon)}(z). \quad (80)$$

Proposition B.6 (Parity changes the shell lattice, not the orientation). *For both elliptic embedding phases,*

$$W_{\text{tp}}^{(\epsilon)}(z^{-1}) = -z(\mathfrak{o}_{G,T} \otimes d_S)W_{\text{tp}}^{(\epsilon)}(z). \quad (81)$$

Proof. One has $W_+(z^{-1}) = -zW_+(z)$. Direct inversion of (78) and (79), using $d_S(Q) = Q^{-1}$, gives $W_-^{(\epsilon)}(z^{-1}) = z d_S W_-^{(\epsilon)}(z)$. The orientation line has eigenvalue +1 on the split projector and -1 on the elliptic projector, so the identities combine as claimed. $\square$

The saturated PGL_2 series and the two SL_2 phase series are distinct rational functions supported on different shell semigroups. Their common inversion line is therefore not determined by the shell lattice.

B.2 The U_2 connected-center test

Proposition B.7 (The connected center forces root orientation). *Let E/F be unramified quadratic and let U_2 be the quasi-split unitary group of an isotropic hermitian plane. Then*

$$U_2^{\text{der}} = SU_2 \simeq SL_2, \quad U_2^{\text{ad}} \simeq PGL_2,$$

and its norm-one connected center acts trivially on the reduced building. For central-neutral spherical families, reciprocity is controlled by

$$\mathfrak{o}_{U_2,T} = \det(X_*(T/Z(U_2)^\circ) \otimes \mathbf{Q}_\ell),$$

whereas replacing this line by $\det(X_(T) \otimes \mathbf{Q}_\ell)$ gives a false identity in the split-root specialization.*

Proof. The central isogeny $Z(U_2)^\circ \times SU_2 \rightarrow U_2$ is the rank-two instance of the center-derived-group decomposition. The adjoint image is type-preserving; equivalently, the connecting map for $1 \rightarrow U_1 \rightarrow U_2 \rightarrow PGL_2 \rightarrow 1$ lands in $H^1(F, U_1) = F^\times / \text{Nm}(E^\times)$ and detects valuation parity. Put $\mathfrak{o}_Z = \det(X_*(Z(U_2)^\circ) \otimes \mathbf{Q}_\ell)$. The rational exact sequence of cocharacter spaces gives

$$\det(X_*(T) \otimes \mathbf{Q}_\ell) = \mathfrak{o}_Z \otimes \mathfrak{o}_{U_2,T}.$$

For a central-neutral split-root family the adjoint series is $W_+(z) = 1/(1 - z)$ and satisfies $W_+(z^{-1}) = -zW_+(z)$. The nontrivial central sign $\mathfrak{o}_Z$ is invisible to both support and building geometry, so inserting it would contradict this identity. Hence the connected-center direction must be deleted. $\square$

B.3 Local classification and the unified theorem

The saturated interpolation, with $\mathfrak{o} = \mathfrak{o}_{G,T}$, is

$$W_{\text{sat}}(z) = \frac{1 - \mathfrak{o}Qz}{(1 - z)(1 - Qz)}, \quad W_{\text{sat}}(z^{-1}) = -z(\mathfrak{o} \otimes d_S)W_{\text{sat}}(z). \quad (82)$$

Theorem B.8 (Local orbital shell classification). *Let (G, K) and γ_m satisfy Definitions 2.2 and 2.3. The Haar orbital coefficients of $\mathbf{1}_K$ are the Frobenius specializations of either*

$$I_{G,m}^{\text{sat}} = 1 + \sum_{r=1}^m (\mathbb{L}_\ell - \mathfrak{o}_{G,T}) \mathbb{L}_\ell^{r-1}, \quad (83)$$

or the type-preserving formulas in Proposition B.5, according to the type image and elliptic phase. This is an equality of local orbit-summed masses. It is a relative orbital push-forward only in the represented saturated PGL_2 family of Theorem A.

Proof. Central neutrality pulls the support condition back from the adjoint group. Lemma B.3 gives the exhaustive allowed radii and phases, while Lemma B.4 gives the core and every positive orbit-summed shell mass. Summing them proves the formulas and their Frobenius specialization. $\square$

Theorem B.9 (Unramified rank-one orientation reciprocity). *Under the hypotheses of Theorem B.8, let W_G be the saturated or type-preserving interpolation selected by the type image and elliptic phase. In Λ_S one has*

$$W_G(z^{-1}) = -z (\mathfrak{o}_{G,T} \otimes d_S) W_G(z), \quad \mathfrak{o}_{G,T} = \det(X_*(T/Z(G)^\circ) \otimes \mathbf{Q}_\ell). \quad (84)$$

The multiplier is independent of the central isogeny, vertex-color lattice, and elliptic embedding phase, although the generating functions differ.

Proof. The classification theorem identifies every local coefficient with the Frobenius trace of the displayed interpolation. The saturated identity is (82); the type-preserving identity is Proposition B.6. Finite central isogenies do not change the rational root cocharacter space, and Proposition B.7 deletes the connected-center direction. This proves the formula in every classified case. $\square$

Corollary B.10 (The Kummer correction is structural). *The line $\mathfrak{o}_{G,T}$ is the sign summand of the quadratic root cameral cover. On a chart $v^2 = u$ it is the Kummer local system K_u , and every allowed positive root shell contains the factor $(\mathbb{L}_\ell - \mathfrak{o}_{G,T}) \mathbb{L}_\ell^{r-1}$ before summation. In the saturated PGL_2 case this factor is represented by $\mathcal{Q}_r$; in all other cases it denotes the orbit-summed interpolation. If the cameral cover is nontrivial, the line is not Tate.*

Proof. The deck involution acts by -1 on the rank-one root cocharacter lattice, so its determinant is the sign summand. The represented assertion is Theorem 4.1; the general trace statement is Lemma B.4. Nontrivial geometric monodromy excludes a constant Tate twist, as in Proposition 1.1. $\square$

C Literature comparison, limitations, and outlook

C.1 Comparison and scope of claim

The numerical input and the relative construction have different roles. Kottwitz's split orbital-integral formula [12, §5.8, (5.8.4)] gives q^m , while his unramified elliptic fixed-vertex count [12, §5.9, (5.9.7)] gives $1 + (q+1)(1 + \cdots + q^{m-1})$. Under our normalization $\text{vol}_G(K) = \text{vol}_T(T(F)_{\text{max, cpt}}) = 1$, the elliptic quotient weights are torus-orbit cardinalities; in the split case they are the compact-unit orbit cardinalities in a translation fundamental section. Lemma B.4 proves these measure identifications, so the quoted sphere and tube counts give precisely these local coefficients. The parity restriction for SL_2 is applied afterwards, with the same torus normalization in both phases.

The relative assertions are the arbitrary-ring stabilizer and fppf quotient calculation in Lemma 5.4, representability in Theorem 5.5, and descent of the determinant line in the residual tower. Work on affine-Springer geometry and fundamental domains provides related background [10, 2, 3, 4]; we make no theorem-by-theorem priority comparison with those works here. Motivic interpolation and transfer are established by [11, 5], and Kummer local systems occur in [7]. The Verdier partner on our finite discrete target follows from standard six-functor duality for the constructed structure morphism. Its role is to record the dual of this specific relative geometry.

The theorem proved here is narrower. It is the componentwise identification of the compact positive-loop orbit sheaf in the universal quadratic PGL_2 family with the single smooth quotient scheme $\mathcal{Q}_r$ over the whole parameter base, together with its enhanced determinant-line descent. The termwise equality of represented shells, Haar masses, and Frobenius traces is stronger than a numerical or virtual interpolation, but it is proved only for this family. The absolute- A_1 theorem makes a different contribution: it shows that the same root-orientation line survives the SL_2 change of shell lattice and that the U_2 center test forces deletion of $X_*(Z(G)^\circ)$. Its general coefficients are local orbit-summed interpolations. These statements specify the scope of the paper; they are not accompanied by a claim that no equivalent formulation exists elsewhere.

Stanley reciprocity and cellular Verdier duality are established tools [17, 8, 1]. The relation between affine-Springer point counting and orbital integrals, the SL_2 vertex-color parity, and the exterior/Poincaré description of torus cohomology are likewise used as standard input. Their role here is to identify and test the height-graded shadow of the represented orbital family. The paper does not claim a new general cellular-duality or torus-duality formalism.

C.2 Exact limitations

Theorem A concerns the equal-characteristic loop family over U_k . It represents each compact $L^+\mathbb{T}$ orbit, but it does not represent the whole affine-Springer locus X_m , the quotient stack $[T \backslash X_m]$, or closures and incidence relations among all its strata. The bookkeeping union $\mathcal{S}_m^{\text{orb}}$ contains one compact fundamental shell per local double coset and is not a global decomposition theorem.

Theorem B is complete only for Definitions 2.2 and 2.3. It excludes ramified tori, anisotropic inner forms, nonspherical test functions, nonstandard depth/fixed-set relations, and relative rank-one groups of non- A_1 absolute type such as quasi-split U_3 . Central neutrality is essential. Outside the saturated PGL_2 case, no relative fppf orbit scheme or derived orbital push-forward has been constructed.

Finally, Theorem 7.1 has a discrete finite target. It is object-level but contains no cross-shell incidence. The two-dimensional cone in Theorem 7.3 appears only after cohomological associated graded, de-Tating, and passage to a rational K_0 character. There is no infinite cellular object, completed Rees module, or convolution theorem hidden in that statement.

C.3 Outlook: the remaining bridge

The next geometric problem is deliberately narrower than a general tropical realization. For the same quadratic PGL_2 family, one should construct actual transition morphisms among the represented shells whose incidence target is the height cone

$$C = \{(m, s) \in \mathbf{N}^2 : 0 \leq s \leq m\}, \quad s = m - r + 1.$$

The desired refinement would upgrade the present discrete functors to an incidence-bearing pair and prove, before taking K_0 ,

$$\mathbb{D}_{\text{cell}} \text{Sk}! \simeq \text{Sk}_* \mathbb{D}.$$

The split tame product charts $\mathcal{Q}_r \times_U \tilde{U} \simeq \mathbb{G}_m \times \mathbb{A}^{r-1}$ give local vertical duality, but they do not yet provide the horizontal transition maps. Constructing those maps is the first unresolved step.

In this limited sense, the present result isolates a rank-one geometric fragment of a possible correspondence-level tropical transport theory.

D Conclusion

The paper proves two results of different strength. First, the universal quadratic PGL_2 family has a componentwise geometric model: $\mathcal{Q}_r$ represents the compact positive-loop orbit of $[\mathcal{A}_r]$, and its finite-field points give exactly the Haar mass of the corresponding spherical double coset. The residual vector-torsor tower identifies every compact shell coefficient with the torus Frobenius bimodule and propagates the non-Tate determinant line $\mathcal{K} = \det(X_*(\mathbb{T}) \otimes \mathbb{Q}_\ell)$ through all depths. The finite labelled-shell map supplies a genuine, if discrete, Verdier partner.

Second, classification and orbit-summed measure transfer extend the reciprocity multiplier to every central-neutral standard unramified absolute- A_1 family. The SL_2 parity test changes the shell lattice while leaving the multiplier fixed; the U_2 center test forces the formula

$$\mathfrak{o}_{G,T} = \det(X_*(T/Z(G)^\circ) \otimes \mathbb{Q}_\ell).$$

Thus the Kummer line in the represented PGL_2 family is the rank-one realization of root orientation, not an accidental numerator in one rational function.

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